Premium or Penalty? Consumer Inference from the Sponsored Label

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Abstract

The "Sponsored" label can make a listing less attractive to consumers in some markets and more attractive in others. I develop a model in which sponsored placement serves both to signal quality and to provide visibility to sellers passed over by the organic algorithm. When both forces operate, consumers favor organic listings under a precise algorithm but sponsored listings under a noisy one. Algorithm noise also shapes the quality distribution of listings shown to consumers: greater noise improves the sponsored pool but shifts the overall quality distribution downward in the sense of first-order stochastic dominance. When choosing algorithm noise, the platform has an incentive to monetize the sponsored label's informational value. This incentive has policy consequences: mandatory sponsorship disclosure can backfire by inducing greater algorithm noise.

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1 Introduction

Consumers routinely face a choice between organic listings and listings promoted through advertising. The "Sponsored" label identifies paid placement, a major source of platform revenue; Amazon alone earned $68.6 billion from advertising services in 2025 (Amazon.com, Inc., 2026). Consumer responses to sponsorship vary across settings. Joo et al. (2024) find a penalty for top-position sponsored listings, whereas Sahni and Nair (2020a) find that disclosing sponsorship increases calls to the advertised restaurant. Abhishek et al. (2025) further show that sponsored listings underperform the organic listings they displace in electronics but outperform them in clothing and footwear. Together, these findings suggest that sponsorship can carry either a *penalty* or a *premium*.

I explain this contrast through an overlooked informational feature of sponsored search: when consumers see a sponsored listing, they do not observe whether the seller would have received organic display had it not advertised. Consumers must therefore infer *why* the seller chose to advertise. A seller passed over by the algorithm has a stronger incentive to advertise because sponsored placement restores the display the algorithm withheld. At the same time, higher-quality sellers obtain greater returns from consumer clicks and are therefore more willing to pay for sponsored placement. Sponsorship thus combines two opposing selection forces: the *bypassing incentive*, which draws in sellers passed over by the organic algorithm, and the *signaling incentive*, through which higher-quality sellers pay more for sponsored placement.

I first show how these incentives jointly shape the label's meaning in a simple environment with binary quality and a posted sponsorship fee. I then extend the model to continuous quality and endogenize the sponsorship fee through an auction. This richer environment allows me to study how algorithm noise affects the overall quality distribution of listings shown to consumers. Finally, I allow the platform to choose algorithm noise endogenously and examine the implications for mandatory sponsorship disclosure.

The first result explains why the same sponsored label can carry a penalty in some markets and a premium in others. Algorithm noise, together with the sponsorship fee or the mass of sponsored listings, determines what consumers infer from the label. Bypassing a precise algorithm is more suspicious than bypassing a noisy one. In the selective-bypass equilibrium, the sponsored pool improves relative to the organic pool as noise rises, generating a unique penalty-to-premium

threshold.

The second result shows how algorithm noise shapes the quality distribution of listings shown to consumers. Holding the mass of sponsored listings fixed, greater noise improves the sponsored pool yet shifts the overall quality distribution downward in the sense of first-order stochastic dominance. Thus, a better sponsored pool need not imply a better overall quality distribution.

The third result concerns mandatory sponsorship disclosure when the platform chooses algorithm noise itself. The platform trades advertising revenue per participant against participation, with greater sensitivity to expected click surplus leading to weakly less algorithm noise.[1] At a fixed algorithm, disclosure weakly raises click surplus and participation. It also lets the platform monetize improvements in the sponsored pool through a higher sponsorship fee. When this revenue effect dominates the change in disclosure's relative participation gain, disclosure makes noisier algorithms relatively more profitable and can induce greater noise. The resulting algorithm adjustment can offset the information gain and leave consumers worse off. I also derive a sufficient condition for disclosure to benefit consumers using observable click-through rates across regimes, without specifying a functional form for the click-cost distribution. The statistic quantifies how far restrictions on that distribution can be relaxed while still ensuring a consumer benefit.

1.1 Related Literature

This paper explains heterogeneity in consumer responses to sponsored labels through what consumers infer from the decision to advertise. Joo et al. (2024), Sahni and Nair (2020a,b), and Abhishek et al. (2025) examine how consumers respond to sponsorship and how sponsored listings perform relative to organic alternatives. Hui et al. (2022) show that making sponsorship disclosure more salient reduces spending on sponsored listings, holding the listings themselves unchanged. The model developed here interprets favorable and unfavorable responses to sponsorship as outcomes of consumer inference about sellers' advertising motives. Because those motives depend on opportunities for organic display, the same label can carry different implications across advertising environments.

The model builds on the literature on advertising as a signal of seller quality. In classic models, higher-quality sellers earn greater returns from advertising, so costly expenditure can select

[1] This tradeoff is at issue in the FTC's ongoing monopolization case against Amazon, which alleges that Amazon knowingly accepted deterioration in search quality because the resulting advertising revenue outweighed losses in transactions (Federal Trade Commission, 2024).

positively on quality (Nelson, 1974; Kihlstrom and Riordan, 1984; Milgrom and Roberts, 1986). Paid-prominence models bring this logic to search markets, allowing consumers to infer relevance or quality from sellers' willingness to pay (Athey and Ellison, 2011; Chen and He, 2011). A related strand considers richer inference from the advertising decision itself. For example, Shin and Yu (2021) show that consumers infer product fit and firm quality from the fact of receiving a targeted advertisement. My model retains positive selection through per-click value and introduces an opposing selection force arising from sellers' incentives to bypass the organic algorithm. When both forces operate, organic precision and the sponsorship fee or the mass of sponsored listings govern their relative strength, determining whether sponsorship conveys favorable or unfavorable information.

The information structure also relates to signaling and countersignaling models in which receivers observe both a costly action and an additional signal of sender quality (Feltovich et al., 2002; Daley and Green, 2014). Favorable outside information can then make costly signaling less valuable for high-quality senders. In my model, sellers observe whether they would receive organic display before deciding whether to advertise, but consumers do not observe this outcome for sponsored sellers. Consumers must therefore infer seller quality from an advertising decision that depends on both quality and the seller's opportunity for organic display.

A related literature examines the interaction between organic and sponsored display. Prior work studies how organic display affects sellers' advertising incentives and bids (Katona and Sarvary, 2010; Xu et al., 2012). Jerath et al. (2011) show that a higher-quality seller can bid less for sponsored prominence yet attract more clicks than its lower-quality rival because some consumers recognize its quality and visit it first regardless of rank. Field experiments on branded search document substantial substitution between paid and organic clicks (Blake et al., 2015; Simonov et al., 2018). Sponsored bidding can also inform organic placement: Long et al. (2022) show how a platform can combine quality information revealed by sellers' bids with its information about consumer fit to improve organic placement. My analysis examines how organic precision shapes the composition of advertisers and consumers' inferences from sponsorship. It connects this selection into advertising to the quality distribution of all displayed listings, showing that a better sponsored pool can coexist with a deterioration in overall displayed quality.

The platform analysis contributes to research on search diversion and endogenous search quality.

Platforms that earn both advertising fees and transaction commissions face a tradeoff between monetizing paid placement and facilitating transactions. Choi and Mela (2019) examine this tradeoff by jointly modeling consumer behavior and sellers' advertising competition. Moshary (2025) provides experimental evidence that sponsored search can reduce total transactions yet remain profitable because advertising revenue outweighs lost commissions. More broadly, intermediaries' incentives to monetize consumer attention can lead them to distort search or display (Hagiu and Jullien, 2011; De Cornière and Taylor, 2014; De Cornière, 2016; Eliaz and Spiegler, 2011; White, 2013; Janssen et al., 2026a). Related work examines how changes in the search environment interact with platform fees and seller pricing. Jiang and Zou (2020) show that lower consumer search costs can induce a platform to raise its referral fee, potentially increasing equilibrium retail prices. Zhong (2023) studies the interaction between platform search precision and pricing. More directly concerned with sponsored placement, Long and Liu (2024) show that a platform can benefit from making an inferior seller appear more attractive and adjusting organic placement to intensify competition for sponsorship. Closest to this paper, Janssen et al. (2026b) show that a platform allocating sponsored and organic positions using match information can raise the value of sponsored placement by obfuscating organic results. I examine how the organic algorithm shapes heterogeneous sellers' advertising decisions and consumers' inferences from those decisions. Greater noise draws higher-quality sellers passed over by the organic algorithm into sponsored placement, improving the sponsored pool. The platform can monetize consumers' more favorable assessment of sponsored listings through a higher sponsorship fee, creating an incentive to increase noise even at the expense of overall displayed quality.

I also explore the model's implications for mandatory sponsorship disclosure, showing that better-informed consumers can nevertheless be worse off once the platform adjusts its algorithm. Park et al. (2023) find that incentivized reviews on Amazon remain inflated despite disclosure and that this inflation raises sales at consumers' expense. Disclosure can also change the incentives of influencers and firms, generating unintended consequences. Requiring influencers to identify paid recommendations can increase their supply of sponsored content (Fainmesser and Galeotti, 2021) or weaken their incentive to build a reputation for useful advice (Mitchell, 2021). Pei and Mayzlin (2022) show that consumers can benefit more from disclosure of whether an influencer is paid than from disclosure of how much the payment biases the review: the more detailed disclosure

enables firms to purchase favorable reviews while keeping them credible enough to induce purchases. Ershov and Mitchell (2025) document an increase in sponsored content following stricter disclosure regulation. Their model explains how partial disclosure can increase trust in unlabeled posts and thereby raise the return to undisclosed sponsorship. My model identifies a different mechanism through which disclosure can backfire, even at a fixed ad load. At a fixed algorithm, disclosure weakly improves consumers' click decisions. However, it also allows the platform to monetize consumers' more favorable assessment of sponsored listings as organic precision falls. The resulting incentive to increase algorithm noise can lead the platform to reduce organic precision enough to outweigh the informational benefit of disclosure.

The remainder of the paper is organized as follows. Section 2 presents the common environment, and Section 3 studies the binary-quality benchmark with a posted sponsorship fee. Section 4 extends the model to continuous quality and endogenizes the sponsorship fee through an auction. Section 5 endogenizes algorithm noise through consumer participation and analyzes mandatory sponsorship disclosure. Section 6 concludes. Appendix A contains proofs of the main results. Appendices B and C contain the full equilibrium correspondence, model foundations, and extensions.

2 The Environment

This section specifies the information, decisions, and payoffs common to the simplified model in Section 3 and the full model in Section 4.

Sellers and organic display. There is a unit mass of atomless sellers. A seller's quality is $q \in [0,1]$, distributed according to CDF F. Quality is measured in units of the consumer's value from clicking. The seller privately observes q and receives per-click value $\kappa(q) > 0$. I assume that κ is strictly increasing: higher-quality sellers have higher per-click values. This captures the greater returns to customer acquisition emphasized in classic advertising-signaling models (Nelson, 1974; Milgrom and Roberts, 1986). Appendix C.2 provides a click–purchase foundation for this quality index and the increasing per-click value.

The organic algorithm generates an organic-display realization $v \in \{0,1\}$ for each seller, with

$$\mathbb{P}(v = 1 \mid q, \sigma) = p(q, \sigma),$$

where $\sigma > 0$ denotes algorithm noise.[2] The realization $v = 1$ means that the seller is granted organic display; $v = 0$ means that the seller is denied organic display. The function p and the noise level σ are common knowledge. The specifications in Sections 3 and 4 describe how greater algorithm noise reduces the informativeness of organic display about quality. The key informational assumption is that sellers observe both q and v before deciding whether to advertise. This reflects an online marketplace setting in which sellers can observe their products' visibility under the existing organic algorithm before purchasing sponsored placement.

Here, the organic algorithm presents the same search results to all consumers. Each seller is therefore either visible organically to everyone or to no one, making it natural for sellers to observe how the algorithm has sorted them before deciding whether to advertise. Appendix C.3 introduces heterogeneous consumer tastes and personalized search results. Bypassing and quality signaling still shape advertising decisions, while the personalized algorithm makes organic display informative about match value as well as quality, giving consumers an additional reason to prefer organic over sponsored listings. Sponsorship then bypasses both the algorithm's assessment of seller quality and its recommendations tailored to individual consumers.

Advertising and displayed listings. Let $a \in \{0, 1\}$ denote the advertising decision and $\alpha(q, v) \in [0, 1]$ the advertising rate among sellers with (q, v). Each seller has at most one displayed listing.[3] If $a = 1$, the seller receives a sponsored listing regardless of v and forgoes any organic display. If $a = 0$ and $v = 1$, the seller receives an organic listing. If $a = 0$ and $v = 0$, the seller remains undisplayed. Thus v describes the organic display available before advertising, whereas the displayed listing depends on both v and a.

To obtain sponsored placement, a seller pays a posted fee f in Section 3 or wins a uniform-price auction for a fixed mass of sponsored listings in Section 4. Under the posted fee, the seller chooses a directly; under the auction, it submits a bid after observing (q, v), and the allocation determines whether it advertises. The seller's information when bidding is therefore the same information used in the posted-fee advertising decision.

[2] Algorithm noise σ is exogenous until Section 5.

[3] This restriction sets aside dual listing. Jerath et al. (2011) report that, among 402 commercial search keywords on Google, about 40% have no overlap between organic and sponsored results and another 40% have only one overlapping link.

Consumer information and click decisions. Under disclosure, a consumer observes a displayed listing's label $\ell \in \{S, O\}$, where S denotes sponsored and O denotes organic. The consumer does not directly observe seller quality q or the organic-display realization v. An organic label implies $(v, a) = (1, 0)$, whereas a sponsored label reveals $a = 1$ but does not reveal v. Consumers know the algorithm and equilibrium advertising strategies. Let $\beta_\ell = \mathbb{E}[q \mid \ell]$ denote the posterior mean quality conditional on label ℓ; I refer to β_S as the sponsored posterior and β_O as the organic posterior.

The consumer observes its own click cost θ, drawn from a distribution with continuous, strictly increasing CDF Φ on $[0, 1]$. Click costs are independent of seller quality and organic display. The consumer's payoff from clicking a listing of quality q is $q - \theta$; its payoff from not clicking is zero. Conditional on (ℓ, θ), the expected payoff from clicking is therefore $\beta_\ell - \theta$. The optimal decision is to click if and only if $\theta \leq \beta_\ell$. It follows that $\mathbb{P}(\text{click} \mid \ell) = \Phi(\beta_\ell)$.

The consumer side of the model abstracts from competition for clicks and defensive advertising incentives.[4] Consumers evaluate listings independently: clicking one listing does not change the payoff or feasibility of clicking another.

Sponsorship is said to carry a penalty when $\beta_S < \beta_O$ and a premium when $\beta_S > \beta_O$. Because Φ is strictly increasing, these posterior comparisons also determine the ordering of click probabilities. In this model, consumers have no intrinsic preference for or against sponsored listings. A penalty or premium arises entirely from what they infer about seller quality from equilibrium advertising decisions.

Seller payoffs and timing. The mass of consumers viewing each displayed listing is normalized to one until participation is introduced in Section 5. A displayed seller's expected clicks are therefore $\Phi(\beta_\ell)$. At a sponsorship fee f, a seller's expected payoff is

$$\pi(q, v, a; f) = a[\kappa(q)\Phi(\beta_S) - f] + (1 - a)v\kappa(q)\Phi(\beta_O).$$

The net gain from advertising is consequently $\kappa(q)[\Phi(\beta_S) - v\Phi(\beta_O)] - f$. At the posted fee, sellers advertise when this gain is positive, do not advertise when it is negative, and can mix when it is zero. Under the auction, the expression before subtracting f is the seller's willingness to pay. Because

[4] Defensive advertising can be an important reason why sellers advertise even when organic display is available (Simonov et al., 2018). In this paper, I focus on equilibria in which sellers granted organic display do not switch to sponsored placement despite having that option.

each seller is atomless, it takes label posteriors and the clearing fee as given, making truthful bidding optimal in the uniform-price auction.

The timing is as follows. The algorithm and allocation mechanism are fixed before the organic-display realizations are generated. Each seller then observes (q, v) and chooses whether to advertise, or submits its auction bid. Sponsored placement is allocated and listings are displayed. Consumers observe the available label information and their click costs and decide whether to click. In equilibrium, sellers optimize, consumers use the optimal click rule, and on-path posteriors satisfy Bayes' rule. The equilibria studied in Sections 3–5 have positive masses of both sponsored and organic listings.

3 Binary Quality and a Posted Fee

The binary-quality model makes the bypassing and signaling incentives transparent. Throughout this section, quality is binary: $q = 1$ has mass $\mu \in (0, 1)$ and $q = 0$ has mass $1 - \mu$. Write $t \in \{H, L\}$ for a seller's type, with H denoting $q = 1$ and L denoting $q = 0$. Let $\kappa_H = \kappa(1) > \kappa_L = \kappa(0) > 0$, $\alpha_{H,v} = \alpha(1, v)$, and $\alpha_{L,v} = \alpha(0, v)$. Sponsored placement costs the posted fee $f \geq 0$. The organic algorithm is summarized by $p_H(\sigma) = p(1, \sigma)$ and $p_L(\sigma) = p(0, \sigma)$.

Assumption 1 *(Organic algorithm with binary quality). The display probabilities are continuously differentiable and satisfy*

$$0 < p_L(\sigma) < p_H(\sigma) < 1, \qquad p'_H(\sigma) < 0 < p'_L(\sigma), \qquad \lim_{\sigma \downarrow 0}(p_H, p_L) = (1, 0).$$

As $\sigma \to \infty$, organic display becomes uninformative about quality: $\lim_{\sigma\to\infty}[p_H(\sigma) - p_L(\sigma)] = 0$.

Greater algorithm noise makes both organic display and its denial less informative about quality. Before advertising, the posterior means are

$$\beta_{v=1}(\sigma) = \frac{\mu p_H(\sigma)}{\mu p_H(\sigma) + (1-\mu)p_L(\sigma)}, \qquad \beta_{v=0}(\sigma) = \frac{\mu[1 - p_H(\sigma)]}{\mu[1 - p_H(\sigma)] + (1-\mu)[1 - p_L(\sigma)]}.$$

They satisfy $\beta_{v=0} < \mu < \beta_{v=1}$. As σ increases, $\beta_{v=1}$ strictly decreases from one toward μ, while $\beta_{v=0}$ strictly increases from zero toward μ. These posteriors give the average quality of sellers granted and denied organic display, respectively, before sellers decide whether to advertise.

3.1 Two Selection Forces

Holding label posteriors fixed, comparing sellers of the same quality across organic-display realizations isolates the bypassing incentive. Comparing quality types among sellers denied organic display isolates the signaling incentive.

Lemma 1 *(Bypassing and signaling). Given label posteriors* (β_S, β_O)*, let* $\Delta_{t,v} = \kappa_t[\Phi(\beta_S) - v\Phi(\beta_O)] - f$ *denote the net gain from advertising for a seller with* (t, v)*. Then*

$$\Delta_{t,0} - \Delta_{t,1} = \kappa_t \Phi(\beta_O) \geq 0, \tag{1}$$

$$\Delta_{H,0} - \Delta_{L,0} = (\kappa_H - \kappa_L)\Phi(\beta_S) \geq 0. \tag{2}$$

The first inequality is strict whenever $\beta_O > 0$*, and the second whenever* $\beta_S > 0$*.*

Equation (1) is the bypassing incentive. A seller passed over by the organic algorithm remains undisplayed unless it advertises, so sponsored placement allows it to bypass the algorithm at no cost in forgone organic clicks. Because denial is more common among low-quality sellers, this incentive creates negative selection into advertising. Equation (2) is the signaling incentive. Among sellers denied organic display, a higher-quality seller values the same sponsored click probability more and therefore has a stronger willingness to pay. This incentive selects positively on quality within the pool drawn toward sponsorship.

3.2 Selective Bypass Equilibrium: From Penalty to Premium

I first study a *selective-bypass equilibrium*, in which sellers granted organic display retain it, every high-quality seller denied display advertises, and low-quality sellers denied display mix. Let $b = \Phi^{-1}(f/\kappa_L)$, the sponsored posterior at which a low-quality seller denied organic display is indifferent between advertising and remaining undisplayed.

Lemma 2 *(Selective bypass). Fix* $\sigma > 0$*. A selective-bypass equilibrium exists if and only if*

$$\kappa_L \Phi\big(\beta_{v=0}(\sigma)\big) < f < \kappa_L \quad \textit{and} \quad \Phi\big(\beta_{v=1}(\sigma)\big) \geq \frac{f(\kappa_H - \kappa_L)}{\kappa_L \kappa_H}.$$

In this equilibrium, advertising rates are

$$\alpha_{H,1} = \alpha_{L,1} = 0, \qquad \alpha_{H,0} = 1, \qquad \alpha_{L,0} = \frac{\mu[1 - p_H(\sigma)](1-b)}{b(1-\mu)[1 - p_L(\sigma)]} \in (0, 1),$$

where $b = \Phi^{-1}(f/\kappa_L)$. *The label posteriors are* $\beta_S = b$ *and* $\beta_O = \beta_{v=1}(\sigma)$.

The first condition ensures that some, but not all, low-quality sellers denied organic display advertise. Their indifference pins down the sponsored posterior at b. The second is the no-switching condition for high-quality sellers granted organic display, ensuring that they prefer to retain it. Because low-quality sellers have a lower per-click value, this condition also rules out their switching to sponsored placement.

Lemma 2 implies that a selective-bypass equilibrium exists for every $\sigma > 0$ whenever the posted fee lies in the interval

$$\mathcal{F}^{SB} = \left(\kappa_L \Phi(\mu), \min\left\{\kappa_L, \frac{\kappa_L \kappa_H \Phi(\mu)}{\kappa_H - \kappa_L}\right\}\right).$$

This interval is always nonempty. For every $f \in \mathcal{F}^{SB}$, the conditions in Lemma 2 hold for all $\sigma > 0$, allowing algorithm noise to vary while the fee remains fixed.

Proposition 1 *(From penalty to premium).* *For every* $f \in \mathcal{F}^{SB}$, *there is a unique threshold* $\widehat{\sigma}(f) > 0$ *satisfying*

$$\beta_{v=1}\big(\widehat{\sigma}(f)\big) = \Phi^{-1}(f/\kappa_L).$$

In the selective-bypass equilibrium, sponsorship carries a penalty for $\sigma < \widehat{\sigma}(f)$ *and a premium for* $\sigma > \widehat{\sigma}(f)$. *At* $\sigma = \widehat{\sigma}(f)$, *the sponsored and organic posteriors coincide. The threshold* $\widehat{\sigma}(f)$ *is strictly decreasing in* f/κ_L.

When algorithm noise is low, organic display is sufficiently informative that the organic posterior exceeds the sponsored posterior. As noise rises, the organic posterior falls below the sponsored posterior, turning the penalty into a premium. A higher fee raises b and therefore lowers the amount of algorithm noise needed for this reversal.

The sponsored posterior remains fixed at b because low-quality sellers denied organic display must remain indifferent. As more high-quality sellers are denied organic display and advertise, the advertising rate of low-quality sellers denied display increases enough to preserve that posterior. At a posted fee, greater noise therefore expands the mass of sponsored listings. Expected advertising revenue is $R^{SB}(\sigma) = f\mu[1 - p_H(\sigma)]/b$, which is strictly increasing in σ (Corollary A.1).

A sponsorship penalty does not imply that advertising is unprofitable. A high-quality seller denied organic display earns $\kappa_H \Phi(b) - f = f(\kappa_H/\kappa_L - 1) > 0$ from advertising even in the penalty

Table 1: The three equilibrium classes analyzed in the binary-quality model.

Class	Supporting conditions	Who advertises	Label implication
Pure bypass	$0 < f \leq \kappa_L \Phi\big(\beta_{v=0}(\sigma)\big)$	Every seller denied organic display, regardless of quality	Sponsorship carries a penalty at every σ
Selective bypass	$\kappa_L \Phi\big(\beta_{v=0}(\sigma)\big) < f < \kappa_L$ and $\Phi\big(\beta_{v=1}(\sigma)\big) \geq \frac{f(\kappa_H - \kappa_L)}{\kappa_L \kappa_H}$	All high-quality sellers denied display; low-quality sellers denied display mix	A unique penalty-to-premium reversal occurs for $f \in \mathcal{F}^{SB}$
Quality signaling	$\kappa_L \Phi(\mu) < f \leq \kappa_H$	Some or all high-quality sellers advertise	Sponsorship carries a premium at every σ

Notes. Sellers granted organic display remain organic in the first two classes. For $f \leq \kappa_L \Phi(\mu)$, selective bypass carries a penalty at every finite noise level at which it exists. The quality-signaling class includes equilibria in which only denied high-quality sellers advertise, all and only high-quality sellers advertise, or all high-quality sellers advertise while low-quality sellers mix. In Table B.1, pure bypass is case 1, selective bypass is case 2, and quality signaling comprises cases 3, 7, and 8. Appendix B.1 gives the complete correspondence and exact existence conditions.

region, because its alternative is to remain undisplayed.

3.3 Other Equilibria and Multiplicity

The selective-bypass equilibrium is one of several equilibria supported by the two selection forces. The generic equilibrium correspondence with both labels on path contains eight cases. Three rely on self-reinforcing switching by high-quality sellers already displayed organically, whereby additional advertising makes further switching more attractive.[5] These equilibria rely on the binary-quality structure and are less central to the mechanism studied here. Table 1 groups the remaining five into three economic classes.

The classes differ in which incentive governs selection into advertising. Low fees leave bypassing dominant; intermediate fees sustain both incentives; high fees screen out low-quality advertising and recover the familiar signaling logic. The supporting fee ranges overlap, allowing multiplicity for some (f, σ). Within the uniform selective-bypass fee region, multiplicity affects the label sign only below $\widehat{\sigma}(f)$; above the threshold, every equilibrium analyzed in the text implies a sponsorship premium. Figure B.1 in Appendix B.1 maps this structure.

The binary-quality model identifies the source of the posterior reversal, with advertising adjusting through low-quality sellers' mixing decisions. In the selective-bypass equilibrium, all undisplayed

[5] Appendix B.1.3 formalizes their self-reinforcing switching incentives.

sellers have quality zero. The next section introduces continuous quality and an auction, replacing mixing with an advertising cutoff and endogenizing the sponsorship fee. This richer environment allows the analysis to trace how algorithm noise changes quality across sponsored, organic, and undisplayed sellers.

4 Continuous Quality and a Sponsored-Listing Auction

In this section, seller quality has CDF F on $[0,1]$ with a continuous, strictly positive density, and $\kappa(q)$ is continuous. Let $\bar{q} = \int_0^1 q\, dF(q)$ denote mean quality. The platform allocates a fixed mass S of sponsored listings, referred to as the ad load, through a uniform-price auction. Sellers submit bids after observing (q, v). Because each seller is atomless, it takes the sponsorship fee and label posteriors as given. Bidding its willingness to pay is therefore optimal, and the market-clearing sponsorship fee f^* equals the willingness to pay of the marginal advertiser.

Assumption 2 *(Organic algorithm with continuous quality).* *The map $p : [0,1] \times (0,\infty) \to (0,1)$ is C^1 and satisfies:*

(i) *$p(\cdot, \sigma)$ is strictly increasing for every σ;*

(ii) *there is $O \in (0,1)$ such that $\int_0^1 p(q,\sigma)\, dF(q) = O$ for every σ;*

(iii) *for every σ, there is $q^\dagger(\sigma) \in (0,1)$ such that $p_\sigma(q,\sigma) > 0$ for a.e. $q < q^\dagger(\sigma)$ and $p_\sigma(q,\sigma) < 0$ for a.e. $q > q^\dagger(\sigma)$, where $p_\sigma \equiv \partial p/\partial \sigma$;*

(iv) *as $\sigma \downarrow 0$, $p(q,\sigma) \to \mathbf{1}\{q > q_O\}$ for a.e. q, where $q_O = F^{-1}(1-O)$; as $\sigma \to \infty$, $p(q,\sigma) \to O$ pointwise.*

Condition (ii) fixes organic capacity, so noise changes which sellers receive organic display without changing its mass. Condition (iii) specifies a single-crossing change: organic-display probabilities increase below $q^\dagger(\sigma)$ and decrease above it. Together, these conditions imply a cumulative reallocation of organic display toward lower-quality sellers (Lemma A.1). Condition (iv) specifies the limits: a perfectly precise algorithm displays the highest-quality sellers, while an uninformative algorithm grants every seller the same display probability. Example 1 shows that Assumption 2 is satisfied when the platform observes a noisy quality score and grants organic display to sellers whose score exceeds a threshold.

Example 1 *(Organic Display with a Noisy Quality Score).* *Suppose the platform observes a noisy quality score $s = q + \sigma\varepsilon$ and grants organic display if the score exceeds a threshold $\bar{s}(\sigma)$ determined by the requirement that organic capacity equals the given mass O. If the noise distribution has a continuously differentiable, strictly positive density, the induced organic-display probability satisfies Assumption 2; Appendix C.1 provides the verification.*

4.1 Selective Bypass Equilibrium under the Auction

Fix an ad load $S \in (0, 1 - O)$. As in the binary model, each seller can occupy at most one displayed listing: a seller that advertises forgoes any organic display. The equilibrium concept is the same as in Section 3, with the sponsorship fee now determined so that the ad load S is filled.

In a selective-bypass equilibrium, sponsored listings are allocated among sellers denied organic display. Their willingness to pay, $\kappa(q)\Phi(\beta_S)$, is strictly increasing in quality. Advertising therefore follows a cutoff $\widehat{q}(\sigma)$, uniquely determined by

$$\int_{\widehat{q}(\sigma)}^{1} [1 - p(q, \sigma)]\, dF(q) = S. \tag{3}$$

Sellers denied organic display with quality above the cutoff exactly fill the ad load. Bayes' rule and the marginal advertiser's willingness to pay give the label posteriors and sponsorship fee:[6]

$$\beta_S(\sigma) = \frac{1}{S} \int_{\widehat{q}(\sigma)}^{1} q[1 - p(q, \sigma)]\, dF(q), \tag{4}$$

$$\beta_O(\sigma) = \frac{1}{O} \int_{0}^{1} qp(q, \sigma)\, dF(q), \tag{5}$$

$$f^*(\sigma) = \kappa(\widehat{q}(\sigma))\Phi(\beta_S(\sigma)). \tag{6}$$

For this allocation to be an equilibrium, sellers granted organic display must prefer to retain it. Their incremental value of sponsored placement is $\kappa(q)[\Phi(\beta_S) - \Phi(\beta_O)]$. When sponsorship carries a premium, this value is greatest at $q = 1$. When sponsorship carries a penalty, no seller granted organic display would pay a positive fee to switch.

[6] Consumer mass is normalized to one here; Section 5.1 explains the scaling when participation is introduced.

Lemma 3 *(Selective-bypass equilibrium under the auction).* *Fix $\sigma > 0$ and $S \in (0, 1-O)$. If the no-switching condition*[7]

$$\kappa(1)[\Phi(\beta_S(\sigma)) - \Phi(\beta_O(\sigma))] \leq f^*(\sigma) \tag{NS}$$

holds, the allocation, label posteriors, and sponsorship fee in (3)–(6) form a selective-bypass equilibrium. Sellers denied organic display advertise for $q > \widehat{q}(\sigma)$ and do not advertise for $q < \widehat{q}(\sigma)$. Sellers granted organic display do not advertise.

The sponsored posterior can also be written as $\beta_S(\sigma) = \mathbb{E}[q \mid v = 0,\ q \geq \widehat{q}(\sigma)]$. Conditioning on denied organic display captures negative selection from the bypassing incentive, while truncation at the advertising cutoff captures positive selection from the signaling incentive. The auction determines how selective advertising must be to fill the fixed ad load.

Let $I(S) = \{\sigma > 0 : \text{(NS) holds}\}$ denote the selective-bypass existence region. Equations (3)–(6) are well defined for every $\sigma > 0$ and describe a selective-bypass equilibrium for $\sigma \in I(S)$.

A *switching equilibrium* has a positive mass of sellers granted organic display choosing to advertise. If (NS) fails, only switching equilibria exist. Every switching equilibrium has $\beta_S > \beta_O$, so sponsorship carries a premium. Switching and selective-bypass equilibria can coexist for some primitives. Appendix B.2 gives the complete correspondence.

The selective-bypass equilibrium captures how sellers use sponsorship to obtain visibility when organic display is unavailable, while sellers already displayed organically have less to gain from advertising. This pattern is consistent with the descriptive evidence in Choi and Mela (2019) that listings with greater organic strength are less likely to be advertised. For prominent brands facing little advertising competition, paid search can add few clicks beyond those obtained organically (Blake et al., 2015; Simonov et al., 2018).

4.2 Algorithm Noise and Consumer Inference

Greater algorithm noise changes both who enters the auction and what consumers infer from each label. The next lemma characterizes the resulting changes in the advertising cutoff and label posteriors.

[7]With heterogeneous consumer tastes, tailoring organic search results to those tastes raises the value of organic display and weakens sellers' incentives to switch into sponsorship, holding the advertising set and organic-display probabilities fixed; see Appendix C.3.

Lemma 4 *(Algorithm noise and auction composition).* *For every $\sigma > 0$,*

$$\widehat{q}'(\sigma) > 0, \qquad \beta_S'(\sigma) > 0, \qquad \beta_O'(\sigma) < 0.$$

Greater algorithm noise reallocates organic display toward lower-quality sellers. At any fixed interior advertising cutoff, the mass of sellers above the cutoff who are denied organic display increases. Because the ad load is fixed, the advertising cutoff rises. Higher-quality advertisers replace lower-quality advertisers, raising the sponsored posterior, while the organic posterior falls. Unlike the posted-fee model, in which low-quality sellers' mixing keeps the sponsored posterior fixed, the auction allows both posteriors to contribute to the relative improvement of sponsored listings.

Proposition 2 *(A unique penalty-to-premium threshold).*

(i) *There is a unique threshold $\widehat{\sigma}(S)$ in the interior of $I(S)$ such that*

$$\beta_S(\sigma) \leq \beta_O(\sigma) \quad \Longleftrightarrow \quad \sigma \leq \widehat{\sigma}(S) \qquad \text{for every } \sigma \in I(S),$$

where equality holds if and only if $\sigma = \widehat{\sigma}(S)$. The threshold depends only on (F, p, O, S), not on κ or Φ.

(ii) *The threshold $\widehat{\sigma}(S)$ is strictly increasing in ad load.*

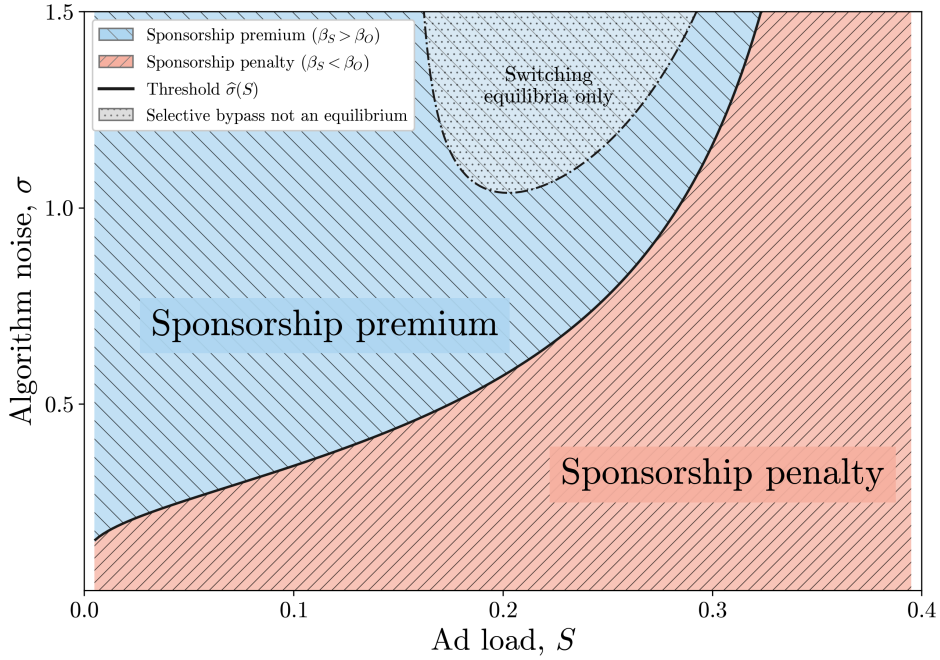
(iii) *For every $\sigma > \widehat{\sigma}(S)$, sponsorship carries a premium in every equilibrium.*

This result extends Proposition 1 to continuous quality and an endogenous sponsorship fee. The common mechanism is that denial of organic display becomes less unfavorable information as noise rises, while willingness to pay selects higher-quality sellers among those denied display.

The threshold depends only on (F, p, O, S) because these objects determine the advertising cutoff and the two label posteriors. The threshold is independent of κ and Φ, but these functions determine the sponsorship fee and the gains from switching in the premium region. At the threshold, sponsored and organic listings receive the same expected clicks. Sellers granted organic display therefore strictly prefer to retain it and avoid the sponsorship fee. By continuity, the no-switching condition also holds in a neighborhood of the threshold, so $\widehat{\sigma}(S) \in \operatorname{int} I(S)$.

A larger ad load admits lower-quality sellers denied organic display into sponsorship. At a fixed algorithm $p(\cdot, \sigma)$, the sponsored posterior falls while the organic posterior is unchanged. Greater algorithm noise is therefore needed for sponsorship to carry a premium, so $\widehat{\sigma}(S)$ increases with S.

Figure 1: Algorithm Noise, Ad Load, and Consumer Inference



Notes. The figure uses the specification in Example 1: $q \sim \text{Unif}[0,1]$, $\varepsilon \sim N(0,1)$, $O = 0.6$, $\Phi(\beta) = \beta$, and $\kappa(q) = 0.02 + 0.98q^2$. The solid curve is $\widehat{\sigma}(S)$, which depends only on (F, p, O, S). Below it, the selective-bypass equilibrium carries a penalty; a switching equilibrium with a premium may coexist. Above it, every equilibrium carries a premium. The dash-dotted no-switching boundary encloses the dotted region where (NS) fails and only switching equilibria exist.

Part (iii) follows because every switching equilibrium carries a premium. Above the threshold, the sign is therefore the same across equilibria. Below it, a selective-bypass equilibrium carries a penalty but may coexist with a switching equilibrium that carries a premium. Figure 1 illustrates these regions in the (S, σ) plane.

4.3 The Quality Distribution of Displayed Listings

Greater algorithm noise improves the quality of sponsored listings but lowers the quality of organic listings. Does the improvement in sponsored quality offset the deterioration in organic quality? Answering this question requires tracing how noise changes which sellers are displayed at all. I therefore examine the quality distribution across all displayed listings, holding both organic capacity and ad load fixed.

Let $\mathcal{M}_\sigma$ denote the quality distribution of displayed listings in the selective-bypass equilibrium. For any Borel set $B \subseteq [0,1]$, $\mathcal{M}_\sigma(B)$ is the probability that a listing drawn uniformly from all

displayed listings has quality in B:

$$\mathcal{M}_\sigma(B) = \frac{1}{O+S}\int_B [p(q,\sigma) + \mathbf{1}\{q \geq \widehat{q}(\sigma)\}[1-p(q,\sigma)]]\, dF(q).$$

The two terms in brackets account for organic and sponsored display, respectively, and $O+S$ is the total mass of displayed listings. Its mean, mean displayed quality, is

$$\bar{q}_{\text{disp}}(\sigma) = \int q\, d\mathcal{M}_\sigma(q) = \frac{O\beta_O(\sigma) + S\beta_S(\sigma)}{O+S}.$$

The next result establishes deterioration of the entire distribution.

Proposition 3 *(Algorithm noise and the quality distribution of displayed listings).* *For every $\sigma_1, \sigma_2 \in I(S)$ with $0 < \sigma_1 < \sigma_2$, the distribution $\mathcal{M}_{\sigma_1}$ strictly first-order stochastically dominates $\mathcal{M}_{\sigma_2}$.*

Greater algorithm noise worsens displayed quality despite improving the sponsored posterior. Sellers above the advertising cutoff are displayed regardless of their organic-display realization: they receive organic display when $v = 1$ and advertise when $v = 0$. Below the cutoff, display depends on the organic algorithm. Greater noise reallocates organic display toward lower-quality sellers, while the rising advertising cutoff leaves some intermediate-quality sellers denied organic display without sponsored placement. The resulting deterioration reduces the expectation of every strictly increasing function of displayed quality.

Proposition 3 identifies a cost of greater algorithm noise: consumers face a worse distribution of displayed listings even as sponsored listings attract more clicks. Without label information, this deterioration reduces the expected surplus from clicking. With labels, the effect on click surplus also depends on how consumers adjust clicks across the two labels. Greater algorithm noise therefore makes consumers more likely to click a sponsored listing, even as it worsens the overall quality distribution of displayed listings. The next section lets participation respond to expected click surplus and studies algorithm choice and sponsorship disclosure.

5 Platform Choice and Sponsorship Disclosure

I now let the platform choose algorithm noise, with consumer participation responding to expected click surplus. I then examine the implications for mandatory sponsorship disclosure.

5.1 Consumer Participation and Optimal Algorithm Noise

The platform commits to algorithm noise $\sigma \in \Sigma = [\sigma_{\min}, \sigma_{\max}] \subset I(S)$, where Σ is compact and the selective-bypass equilibrium is selected throughout.[8] Organic capacity O and ad load S remain fixed. There is a unit mass of potential consumers. Consumers know the algorithm when deciding whether to participate, before organic-display realizations and their click costs are observed. Participants then observe listing labels and their click costs and decide whether to click.

For a listing with posterior mean quality b, expected click surplus is $\omega(b) = \mathbb{E}_\theta[(b-\theta)^+] = \int_0^b \Phi(t)\,dt$. The expected total click surplus of a participating consumer is

$$W(\sigma) = O\omega(\beta_O(\sigma)) + S\omega(\beta_S(\sigma)). \tag{7}$$

This measure averages over click costs and label information before either is observed and adds surplus across all displayed listings. Appendix C.4 derives participation from consumers comparing this surplus with an outside option.

Assumption 3 *(Consumer participation).* *For each λ, let $N_\lambda(W) \in (0,1]$ be the mass of consumers who participate when their expected total click surplus is W. On the relevant range of positive surplus:*

(i) *N_λ is strictly increasing in W.*

(ii) *For $\lambda_2 > \lambda_1$ and $W_1 > W_2$,*

$$\frac{N_{\lambda_2}(W_2)}{N_{\lambda_2}(W_1)} < \frac{N_{\lambda_1}(W_2)}{N_{\lambda_1}(W_1)}. \tag{PS}$$

(iii) *N_λ is continuous in W.*

Condition (PS) says that the same reduction in expected click surplus causes a larger proportional

[8] Appendix B.2.3 provides a sufficient condition under which the selective-bypass equilibrium is unique at every $\sigma \in \Sigma$.

participation loss when λ is higher. Thus λ measures *participation sensitivity*. The condition does not restrict how participation at a fixed W varies with λ.

Advertising revenue per participating consumer is

$$R(\sigma) = Sf^*(\sigma) = S\kappa(\widehat{q}(\sigma))\Phi(\beta_S(\sigma)).$$

It strictly increases with noise because both $\widehat{q}$ and β_S increase. Participation scales all sellers' returns and willingness to pay by the same factor, leaving the advertising allocation, label posteriors, and no-switching condition unchanged. The platform solves

$$\max_{\sigma \in \Sigma} N_\lambda(W(\sigma))R(\sigma).$$

Assume this problem has a unique maximizer, denoted by $\sigma^*(\lambda)$.[9]

Lemma 5 *(Participation sensitivity and optimal algorithm noise).* *If $\lambda_2 > \lambda_1$, then $\sigma^*(\lambda_2) \leq \sigma^*(\lambda_1)$ and $W(\sigma^*(\lambda_2)) \geq W(\sigma^*(\lambda_1))$.*

The platform trades off advertising revenue per participant against the number of consumers who participate. Greater noise raises revenue per participant, while greater expected click surplus attracts more consumers. Higher participation sensitivity makes a given surplus loss more costly through lost participation, leading to weakly less noise and weakly greater click surplus.

The revenue incentive is consistent with the field evidence in Moshary (2025): advertising revenue can make worse marketplace outcomes profitable for the platform. Appendix C.5 adds transaction commissions and states when participation sensitivity still lowers optimal noise and when commissions themselves favor a more precise algorithm.

5.2 Mandatory Sponsorship Disclosure

Federal Trade Commission (2015) calls for clear and prominent disclosure when consumers might otherwise be misled about the commercial nature of advertising. Disclosure helps consumers make informed click decisions, but it also changes both participation and the platform's return to algorithm precision.

[9]Without uniqueness, the participation-sensitivity comparison applies to the optimizer correspondence.

Write $W_D(\sigma) \equiv W(\sigma)$ for click surplus under disclosure, where consumers observe sponsored and organic labels. Under no disclosure, they know that displayed listings include both types but cannot distinguish them, using the pooled posterior $\bar{q}_{\text{disp}}(\sigma)$. Sellers' information, the auction, capacities O, S, and the feasible set Σ are unchanged. A seller granted organic display gains no additional clicks from advertising without labels; among denied sellers, willingness to pay remains strictly increasing in quality. The same cutoff $\widehat{q}(\sigma)$ therefore clears the auction.[10] The fee and advertising revenue per participant are $f_{ND}(\sigma) = \kappa(\widehat{q}(\sigma))\Phi(\bar{q}_{\text{disp}}(\sigma))$ and $R_{ND}(\sigma) = Sf_{ND}(\sigma)$.

Expected total click surplus per participant under no disclosure is

$$W_{ND}(\sigma) = (O + S)\omega(\bar{q}_{\text{disp}}(\sigma)).$$

The same function $N_\lambda(W)$ governs participation in both regimes, so participation can differ even at a fixed algorithm. Write $\sigma_D^*(\lambda) \equiv \sigma^*(\lambda)$ for the disclosure optimum above. Assume that $N_\lambda(W_{ND}(\sigma))R_{ND}(\sigma)$ also has a unique maximizer, denoted by $\sigma_{ND}^*(\lambda)$. Since W_{ND} strictly decreases with noise (Appendix A.4, "Click surplus and algorithm noise"), (PS) also makes σ_{ND}^* weakly decreasing in λ.[11] For brevity, write $\widehat{\sigma} = \widehat{\sigma}(S)$.

Proposition 4 *(Disclosure and algorithm noise)***.**

(i) *For any fixed $\sigma \in \Sigma$, removing the label leaves the advertising allocation and displayed-quality distribution unchanged. The sponsorship fees per participating consumer satisfy $f^*(\sigma) \gtrless f_{ND}(\sigma)$ iff $\sigma \gtrless \widehat{\sigma}$, with equality iff $\sigma = \widehat{\sigma}$.*

(ii) *For any fixed $\sigma \in \Sigma$, disclosure weakly raises expected total click surplus per participant, $W_D(\sigma) \geq W_{ND}(\sigma)$, and participation. Both inequalities are strict if $\sigma \neq \widehat{\sigma}$.*

(iii) *If the following ratio is weakly increasing in σ on Σ, then $\sigma_D^*(\lambda) \geq \sigma_{ND}^*(\lambda)$:*

$$\frac{N_\lambda(W_D(\sigma))}{N_\lambda(W_{ND}(\sigma))} \frac{\Phi(\beta_S(\sigma))}{\Phi(\bar{q}_{\text{disp}}(\sigma))}. \tag{8}$$

Disclosure improves click decisions: in the penalty region, consumers with click costs between β_S and $\bar{q}_{\text{disp}}$ avoid sponsored listings, while those with costs between $\bar{q}_{\text{disp}}$ and β_O make additional

[10] No switching equilibrium exists under no disclosure; the selective-bypass outcome is unique at each σ.

[11] The proof of Lemma 5 also establishes this no-disclosure comparison.

organic clicks. These gains also make consumers more willing to participate. The fee comparison in part (i) is per participant; the actual fee also reflects the participating mass.

The condition in part (iii) compares the participation gain and the advertising-revenue gain from disclosure. The second ratio in (8) strictly increases with noise: sponsored quality rises while pooled quality falls. Disclosure makes sponsored clicks more responsive to advertiser quality, allowing the platform to raise the sponsorship fee per participant by increasing noise and improving the sponsored pool. The first ratio can fall when noise reduces the value of distinguishing the labels. Part (iii) requires the revenue effect to offset any such loss of the relative participation gain. A sufficiently strong participation response can instead favor a more precise algorithm.

For example, let $\Phi(\theta) = \theta$ and $N_\lambda(W) = (W/\bar{W})^\lambda$, where $\lambda > 0$ and the fixed $\bar{W}$ exceeds all relevant surplus values. This family satisfies Assumption 3, with λ equal to the elasticity of participation with respect to surplus. A sufficient condition for part (iii) is $S\lambda^2 \leq O(1 + 2\lambda)$ (Appendix A.4). Thus the revenue effect dominates when participation is not too sensitive; the bound permits greater sensitivity when organic capacity is larger relative to ad load.

Participation responds to expected click surplus in both regimes and can limit or reverse this algorithm adjustment. Fix λ and omit it from the notation. The net effect of disclosure on click surplus per participant is

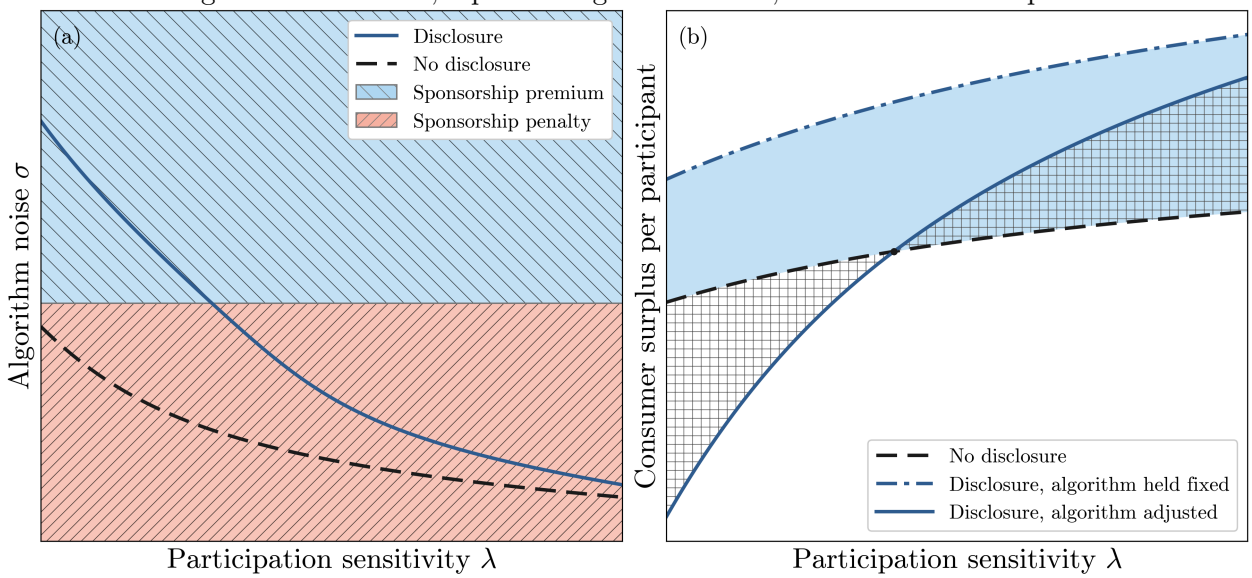
$$W_D(\sigma_D^*) - W_{ND}(\sigma_{ND}^*) = \underbrace{W_D(\sigma_{ND}^*) - W_{ND}(\sigma_{ND}^*)}_{\text{informational benefit}} + \underbrace{W_D(\sigma_D^*) - W_D(\sigma_{ND}^*)}_{\text{platform response}}. \tag{9}$$

The first term is nonnegative. When the condition in part (iii) holds and both optima lie at or below $\widehat{\sigma}$, the second is nonpositive because W_D decreases with noise in the penalty region (Appendix A.4). If the loss from the platform response exceeds the informational benefit, disclosure leaves consumers worse off and reduces participation.

Figure 2 illustrates how the platform response can outweigh the informational benefit. Panel (a) compares optimal noise as participation sensitivity rises. Panel (b) shows the informational benefit, platform response, and resulting net effect.

The net gain need not be monotone in λ. If participation becomes sufficiently sensitive to surplus, however, each regime's optimum approaches its highest attainable click surplus. Disclosure then

Figure 2: Disclosure, Optimal Algorithm Noise, and Consumer Surplus



Notes. Primitives follow Figure 1, with $S = 0.06$, $\Sigma = [0.05, 0.55] \subset I(S)$, and $N_\lambda(W) = (3W)^\lambda$. Panels (a) and (b) use $\lambda \in [6.5, 18.5]$ and $\lambda \in [12.5, 18.5]$, respectively. The selective-bypass equilibrium is selected throughout; switching equilibria coexist at some low noise levels, below all plotted optima. The mandate binds throughout panel (b): the platform would otherwise choose no disclosure (Remark 1). Blue shading in (b) shows the informational benefit at σ^*_{ND}. The difference between the two disclosure curves is the platform response; the grid pattern shows the net effect relative to no disclosure. The black point marks the sign change of this net effect. Relative to no disclosure, surplus per participant changes from -1.60% to $+1.00\%$ across panel (b), crossing zero at $\lambda \approx 14.85$.

benefits consumers whenever its highest attainable surplus is strictly greater.[12]

A mandate matters only if the platform would otherwise withhold labels. The following remark gives sufficient conditions for this voluntary choice.

Remark 1 *(Voluntary disclosure).* *The platform strictly prefers disclosure if* $\sigma^*_{ND}(\lambda) > \widehat{\sigma}$. *It strictly prefers no disclosure if, at* $\sigma = \sigma^*_D(\lambda)$,

$$\frac{N_\lambda(W_D(\sigma))}{N_\lambda(W_{ND}(\sigma))} < \frac{\Phi(\bar{q}_{\text{disp}}(\sigma))}{\Phi(\beta_S(\sigma))}.$$

In the first case, disclosure raises both revenue per participant and participation at the no-disclosure optimum. In the second, its participation gain is too small to offset the revenue loss at the disclosure optimum, so a mandate binds. A sponsorship penalty alone does not establish that the platform would prefer no disclosure.

[12] A sufficient condition is $N_\lambda(W_2)/N_\lambda(W_1) \to 0$ as $\lambda \to \infty$ for every $W_1 > W_2 > 0$. Since W_{ND} is maximized at $\sigma_{\min}$, $\sigma_{\min} \neq \widehat{\sigma}$ guarantees a strict gap between the two regimes' highest attainable surplus values. The power family above satisfies this limiting condition; the condition in part (iii) need not hold for arbitrarily large λ. Appendix A.4 proves the limiting claim.

5.3 Click-Through Rates and Consumer Surplus

The platform's algorithm response can overturn the consumer benefit of disclosure. This gives policymakers a reason to consider oversight of organic precision alongside disclosure requirements.[13] However, organic precision is difficult to observe directly. Researchers can measure click-through rates and the proportion of sponsored listings among displayed results. Within the model, these observables determine the sign of the consumer-welfare effect of disclosure if the click-cost distribution Φ is known. In practice, Φ may be difficult to estimate precisely. I therefore construct a threshold from click-through rates across the two regimes and the proportion of sponsored listings. This threshold identifies shape restrictions on Φ sufficient to establish a consumer benefit without specifying the full distribution.

Maintain the selective-bypass equilibrium and the same primitives across regimes, including the click-cost distribution among participating consumers. Define

$$c_O = \Phi(\beta_O(\sigma_D^*)), \qquad c_S = \Phi(\beta_S(\sigma_D^*)), \qquad c_{ND} = \Phi(\bar{q}_{\text{disp}}(\sigma_{ND}^*)),$$

and write $w_O = O/(O+S)$ and $w_S = S/(O+S)$. Here c_O and c_S are the organic and sponsored click-through rates under disclosure, c_{ND} is the pooled rate under no disclosure, and w_O and w_S are the proportions of organic and sponsored listings among displayed results.

Under the model's click rule, a click-through rate c corresponds to posterior mean quality $\Phi^{-1}(c)$ and expected surplus $\omega(\Phi^{-1}(c))$. Consequently,

$$\Delta W \equiv W_D(\sigma_D^*) - W_{ND}(\sigma_{ND}^*) = (O+S)\big[w_O\omega(\Phi^{-1}(c_O)) + w_S\omega(\Phi^{-1}(c_S)) - \omega(\Phi^{-1}(c_{ND}))\big]. \quad (10)$$

When a mandate binds, this difference measures its effect on expected total click surplus per participant. It includes both the informational benefit and the platform response, regardless of which regime chooses greater noise. Under the outside-option foundation in Appendix C.4, its sign also determines the change in aggregate consumer welfare, including participation and outside options.

If both label-specific rates are below the pooled benchmark, disclosure reduces surplus because

[13] The UK's Fair Ranking requirement for Google provides an example of regulatory oversight of organic ranking criteria (Competition and Markets Authority, 2026).

$\omega(\Phi^{-1}(c))$ is increasing in c. When one rate is higher and the other lower, the comparison depends on how click costs translate these changes into surplus. To characterize this dependence, define the curvature index

$$\mathcal{R}_\Phi(\theta) = 2 - \frac{\Phi(\theta)\Phi''(\theta)}{\Phi'(\theta)^2},$$

where Φ is twice continuously differentiable and $\Phi' > 0$. Log-concavity of Φ implies $\mathcal{R}_\Phi \geq 1$, while concavity implies $\mathcal{R}_\Phi \geq 2$.

Corollary 1 *(Click-through rates and consumer welfare). If $w_O c_O + w_S c_S < c_{ND} < \max\{c_O, c_S\}$, there is a unique $r^* > 1$ satisfying*

$$w_O c_O^{r^*} + w_S c_S^{r^*} = c_{ND}^{r^*}.$$

If Φ is twice continuously differentiable with $\Phi' > 0$ and $\mathcal{R}_\Phi(\theta) \geq r^*$ throughout the interval between $\beta_O(\sigma_D^*)$ and $\beta_S(\sigma_D^*)$, then $\Delta W \geq 0$.

The corollary addresses the case in which disclosure lowers the average click-through rate but raises the rate for one label above the pooled benchmark. The threshold r^* is calculated entirely from click-through rates and the proportion of sponsored listings among displayed results. A lower r^* means that a weaker curvature restriction suffices to guarantee a consumer benefit.

Two familiar restrictions give simple criteria. If Φ is log-concave over the interval between the two disclosure posteriors, then

$$w_O c_O + w_S c_S \geq c_{ND} \quad \Longrightarrow \quad \Delta W \geq 0.$$

A weak increase in the average click-through rate is therefore sufficient for a consumer benefit. If Φ is concave over that interval, the weaker criterion

$$w_O c_O^2 + w_S c_S^2 \geq c_{ND}^2 \quad \Longrightarrow \quad \Delta W \geq 0$$

applies. In particular, when $1 < r^* \leq 2$, concavity establishes a consumer benefit even though the average click-through rate falls. When $r^* > 2$, concavity alone does not guarantee that conclusion.

The criterion provides a conservative basis for concluding that disclosure benefits consumers: within the model, the conclusion holds for every click-cost distribution satisfying the stated shape

restriction. Failure to meet the criterion does not establish consumer harm but calls for further investigation.

6 Concluding Remarks

A sponsored label tells consumers that a seller paid to be displayed. Consumers may therefore use the seller's decision to advertise to infer the quality of the listing. In sponsored search, however, this inference differs from the standard signaling logic of advertising. Sellers advertise not only because high-quality sellers have more to gain from consumer clicks, but also because sponsorship provides an alternative route to display when the organic algorithm passes them over. Advertising therefore combines two opposing forces: being willing to advertise can be good news about quality, while having needed advertising to appear at all can be bad news.

This paper develops a framework that brings these two forces together. Because consumers do not observe where a sponsored listing would have appeared absent sponsorship, the informational content of the sponsored label cannot be understood separately from the organic ranking process. The precision of the organic algorithm is therefore central. When organic display is sufficiently informative about quality, sponsorship can reveal that a seller was unlikely to have appeared organically, making consumers less willing to click. When the signaling force is stronger, the opposite occurs. The same sponsored label can consequently carry either a penalty or a premium, depending on how organic selection shapes the pool of sellers that choose to advertise.

The interaction between organic selection and advertising also has implications beyond consumers' inference from the label. As the organic algorithm becomes noisier, higher-quality sellers that are passed over increasingly enter sponsored placement and can displace lower-quality advertisers. The average quality of the sponsored pool may therefore improve. At the same time, noisier organic selection makes the platform less effective at directing display toward higher-quality sellers, reducing the quality of the marketplace as a whole. An improvement in the observed performance of sponsored listings can thus coexist with a deterioration in the quality of displayed content overall.

These forces also shape the platform's incentives when the organic algorithm is chosen strategically. Consumers participate based on the surplus they expect from their subsequent click decisions, while the platform also values the advertising revenue generated by each participating consumer. Greater

participation sensitivity gives the platform a stronger incentive to preserve organic precision, whereas the revenue effect pushes toward greater noise. As a result, greater participation sensitivity leads the platform to choose weakly less noise, benefiting consumers. Because the algorithm also determines what consumers infer from the sponsored and organic labels, this tradeoff can change whether sponsorship is perceived as a penalty or a premium.

Finally, the analysis shows why the welfare effects of disclosure depend on the platform's response. At a fixed algorithm, disclosure improves consumers' click decisions, thereby weakly increasing both consumer welfare and participation. Once algorithm design is endogenous, however, disclosure also changes the platform's return to organic precision. As precision falls, higher-quality sellers that are passed over enter sponsored placement, improving the sponsored pool and allowing the platform to monetize consumers' more favorable assessment of sponsored listings. When this revenue effect is sufficiently strong relative to any decline in disclosure's participation advantage, the platform chooses weakly greater noise. The resulting loss in consumer welfare can outweigh disclosure's direct informational benefit. Under restrictions on the click-cost distribution, observed click-through rates provide a sufficient condition for identifying when disclosure benefits consumers.

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A Proofs of the Main Results

A.1 Preliminaries

For the continuous-quality model, write $m_0(x,\sigma) = \int_x^1 [1-p(q,\sigma)]\,dF(q)$ for the mass of $v=0$ sellers above x, so that $m_0(\widehat{q}(\sigma),\sigma) = S$.

Lemma A.1 *(Cumulative rotation). Write $\gamma_\sigma(x) = \int_0^x p_\sigma(q,\sigma)\,dF(q)$ for the cumulative change in organic mass below x. Under Assumption 2, for every $\sigma > 0$:*

(a) *$\gamma_\sigma(x) > 0$ for every $x \in (0,1)$ and $\gamma_\sigma(0) = \gamma_\sigma(1) = 0$;*

(b) *for every $x \in [0,1)$,*

$$\int_x^1 (q-x)(-p_\sigma(q,\sigma))\,dF(q) = \int_x^1 \gamma_\sigma(y)\,dy > 0.$$

Proof. Fixed organic capacity gives $\gamma_\sigma(0) = \gamma_\sigma(1) = 0$. The single-crossing sign of p_σ makes γ_σ strictly increase up to $q^\dagger$ and strictly decrease thereafter, proving (a). For (b), integration by parts gives

$$\int_x^1 (q-x)(-p_\sigma)\,dF = -[(q-x)\gamma_\sigma(q)]_x^1 + \int_x^1 \gamma_\sigma(y)\,dy = \int_x^1 \gamma_\sigma(y)\,dy > 0.$$

□

Lemma A.2 *(Ratio comparative statics). Let $\Pi_1, \Pi_2 : \Sigma \to (0,\infty)$ and let $\sigma_i \in \arg\max_{\sigma\in\Sigma} \Pi_i(\sigma)$ for $i = 1,2$.*

(i) *If Π_2/Π_1 is strictly decreasing on Σ, then $\sigma_2 \le \sigma_1$; if strictly increasing, then $\sigma_2 \ge \sigma_1$.*

(ii) *The same conclusions hold if the ratio is only weakly monotone in the stated direction, provided at least one of σ_1, σ_2 is the unique maximizer of its own problem.*

Proof. For the decreasing case, suppose $\sigma_2 > \sigma_1$. Optimality gives $\Pi_2(\sigma_2) \ge \Pi_2(\sigma_1)$ and $\Pi_1(\sigma_1) \ge \Pi_1(\sigma_2)$; all values are positive, so dividing yields $\Pi_2(\sigma_2)/\Pi_1(\sigma_2) \ge \Pi_2(\sigma_1)/\Pi_1(\sigma_1)$, contradicting strict decrease. Under (ii), uniqueness makes one of the two optimality inequalities strict, so the quotient inequality is strict and contradicts weak decrease. The increasing case is symmetric. □

A.2 Binary-Quality Model (Section 3)

Proof of Lemma 1. For either type $t \in \{H, L\}$, $\Delta_{t,0} = \kappa_t \Phi(\beta_S) - f$ and $\Delta_{t,1} = \kappa_t[\Phi(\beta_S) - \Phi(\beta_O)] - f$; subtracting gives $\Delta_{t,0} - \Delta_{t,1} = \kappa_t \, \Phi(\beta_O)$, which is strictly positive whenever $\beta_O > 0$ because $\kappa_t > 0$, Φ is strictly increasing, and $\Phi(0) = 0$. Subtracting across qualities within the $v = 0$ cell, $\Delta_{H,0} - \Delta_{L,0} = (\kappa_H - \kappa_L) \, \Phi(\beta_S)$, strictly positive whenever $\beta_S > 0$ because $\kappa_H > \kappa_L$. □

Proof of Lemma 2. In any selective-bypass profile, $\alpha_{H,0} = 1$, $\alpha_{H,1} = \alpha_{L,1} = 0$, and $\alpha_{L,0} \in (0, 1)$, so both labels occur with positive mass and Bayes' rule gives $\beta_O = \beta_{v=1}(\sigma)$ and

$$\beta_S = \frac{\mu[1 - p_H]}{\mu[1 - p_H] + (1 - \mu)[1 - p_L] \, \alpha_{L,0}},$$

which is strictly decreasing in $\alpha_{L,0}$ with range $(\beta_{v=0}(\sigma), 1)$. Because $(L, 0)$ mixes, it must be indifferent: $\Delta_{L,0} = \kappa_L \Phi(\beta_S) - f = 0$, so $\beta_S = \Phi^{-1}(f/\kappa_L) = b$. By strict monotonicity, the unique $\alpha_{L,0}$ consistent with $\beta_S = b$ is the rate stated in the lemma, and it is interior precisely when $b \in \big(\beta_{v=0}(\sigma), 1\big)$, that is, when $\kappa_L \Phi(\beta_{v=0}(\sigma)) < f < \kappa_L$.

For the remaining cells, using $\Phi(\beta_S) = f/\kappa_L$,

$$\Delta_{H,0} = f\Big(\frac{\kappa_H}{\kappa_L} - 1\Big) > 0, \qquad \Delta_{L,1} = -\kappa_L \, \Phi(\beta_{v=1}) < 0,$$

where the signs use $f > \kappa_L \Phi(\beta_{v=0}) > 0$ and $\beta_{v=1} > \mu > 0$, and

$$\Delta_{H,1} = \kappa_H\Big(\frac{f}{\kappa_L} - \Phi\big(\beta_{v=1}\big)\Big) - f \le 0 \quad \iff \quad \Phi\big(\beta_{v=1}(\sigma)\big) \ge \frac{f(\kappa_H - \kappa_L)}{\kappa_L \kappa_H}.$$

Thus $\alpha_{H,0} = 1$ and $\alpha_{L,1} = 0$ are strictly optimal on the fee interval, while $\alpha_{H,1} = 0$ is optimal if and only if the no-switching condition holds. The selective-bypass profile is therefore an equilibrium exactly under the stated conditions, with $\beta_S = b$ and $\beta_O = \beta_{v=1}(\sigma)$. □

Corollary A.1 *(Advertising revenue in the selective-bypass equilibrium). For every $f \in \mathcal{F}^{SB}$, expected advertising revenue per consumer is*

$$R^{SB}(\sigma) = \frac{f\mu[1 - p_H(\sigma)]}{b}, \qquad b = \Phi^{-1}(f/\kappa_L).$$

Revenue strictly increases in σ.

Proof. Bayes' rule gives an advertiser mass of $\mu(1 - p_H)/b$, so $R^{SB} = f\mu(1 - p_H)/b$. Since $f\mu/b > 0$ is independent of σ and $p_H' < 0$, $(R^{SB})' = -(f\mu/b)p_H' > 0$. □

Proof of Proposition 1. By Lemma 2, $\beta_S = b = \Phi^{-1}(f/\kappa_L)$ is constant in σ, while $\beta_O(\sigma) = \beta_{v=1}(\sigma)$; and because any $f \in \mathcal{F}^{SB}$ sustains this equilibrium at every $\sigma > 0$, the comparison below takes place within the selective-bypass equilibrium as noise varies. Since $\mathcal{F}^{SB} \subseteq (\kappa_L \Phi(\mu), \kappa_L)$, we have $b \in (\mu, 1)$.

By Assumption 1, $\beta_{v=1}(\sigma)$ is continuous and strictly increasing in the ratio $p_H(\sigma)/p_L(\sigma)$, which strictly decreases in σ because $p'_H < 0 < p'_L$; hence $\beta_{v=1}$ is strictly decreasing. As $\sigma \downarrow 0$, $(p_H, p_L) \to (1, 0)$ gives $\beta_{v=1} \to 1$; as $\sigma \to \infty$, $p_H - p_L \to 0$ while p_L, being strictly increasing, is bounded away from zero, so $p_H/p_L \to 1$ and $\beta_{v=1} \to \mu$. Thus $\beta_{v=1}$ maps $(0, \infty)$ bijectively onto $(\mu, 1)$, and since $b \in (\mu, 1)$ there is a unique $\widehat{\sigma}(f)$ with $\beta_{v=1}(\widehat{\sigma}(f)) = b$. Strict monotonicity then yields $\beta_O(\sigma) > b = \beta_S$ for $\sigma < \widehat{\sigma}(f)$, equality at $\sigma = \widehat{\sigma}(f)$, and $\beta_O(\sigma) < \beta_S$ for $\sigma > \widehat{\sigma}(f)$. Finally, $\widehat{\sigma}(f) = \beta_{v=1}^{-1}(\Phi^{-1}(f/\kappa_L))$; since Φ^{-1} is strictly increasing and $\beta_{v=1}^{-1}$, the inverse of a strictly decreasing function, is strictly decreasing, $\widehat{\sigma}(f)$ strictly decreases in f/κ_L. □

A.3 Continuous Quality (Section 4)

Proof of Lemma 3. Since F has a continuous, strictly positive density and p is C^1, the map $(x, \sigma) \mapsto \int_x^1 [1 - p(q, \sigma)]\, dF(q)$ is C^1 with x-derivative $-[1 - p(x, \sigma)]F'(x) < 0$, and it falls from $1 - O > S$ at $x = 0$ (Assumption 2(ii)) to $0 < S$ at $x = 1$. Hence (3) has a unique solution $\widehat{q}(\sigma) \in (0, 1)$, and the implicit function theorem gives $\widehat{q} \in C^1$ (used in the proof of Lemma 4). Because $q \geq \widehat{q}$ on the domain of integration in (4), comparing with (3) yields $\beta_S \geq \widehat{q} > 0$, so $\Phi(\beta_S) \geq \Phi(\widehat{q}) > \Phi(0) = 0$ and, by (6), $f^* > 0$.

A seller with $v = 0$ receives 0 if it remains undisplayed and, by (6), $[\kappa(q) - \kappa(\widehat{q})]\Phi(\beta_S)$ if it advertises. Since κ is strictly increasing and $\Phi(\beta_S) > 0$, advertising is strictly optimal for $q > \widehat{q}$ and strictly suboptimal for $q < \widehat{q}$, with the marginal advertiser $\widehat{q}$ indifferent: the stated rule is optimal, f^* is the marginal advertiser's willingness to pay, and by (3) the advertisers fill the ad load exactly. A seller with $v = 1$ receives $\kappa(q)\Phi(\beta_O)$ from organic display, while advertising forfeits it and yields $\kappa(q)\Phi(\beta_S) - f^*$; not advertising is optimal iff $\kappa(q)[\Phi(\beta_S) - \Phi(\beta_O)] \leq f^*$. If $\Phi(\beta_S) \leq \Phi(\beta_O)$, this holds for every q because $f^* > 0$; otherwise the left side is strictly increasing in q, so the condition for every q reduces to its instance at $q = 1$, which is the no-switching condition (NS).

Finally, the sponsored pool $\{v = 0,\, q \geq \widehat{q}(\sigma)\}$ and the organic pool $\{v = 1\}$ have masses $S > 0$ and $O > 0$, and Bayes' rule on them returns exactly (4) and (5). Sellers optimize, the ad load is

filled, and both labels are on path with consistent posteriors, so (4)–(6) form a selective-bypass equilibrium with the stated advertising rule. □

Proof of Lemma 4. The regularity in Lemma 3 permits differentiation of the capacity and posterior identities. Differentiating capacity and using $\gamma_\sigma(1) = 0$ gives

$$\widehat{q}'(\sigma)[1 - p(\widehat{q}, \sigma)]F'(\widehat{q}) = \int_{\widehat{q}}^{1} (-p_\sigma)\, dF = \gamma_\sigma(\widehat{q}) > 0. \tag{A.1}$$

Using (A.1) when differentiating β_S, and applying Lemma A.1 to both posteriors, yields

$$S\beta_S' = \int_{\widehat{q}}^{1} (q - \widehat{q})(-p_\sigma)\, dF = \int_{\widehat{q}}^{1} \gamma_\sigma(y)\, dy > 0,$$
$$O\beta_O' = \int_{0}^{1} q\, p_\sigma\, dF = -\int_{0}^{1} \gamma_\sigma(y)\, dy < 0.$$

Together with $[1 - p(\widehat{q}, \sigma)]F'(\widehat{q}) > 0$, these identities give all three signs. Adding the two posterior derivatives also gives

$$S\beta_S'(\sigma) + O\beta_O'(\sigma) = -\int_{0}^{\widehat{q}(\sigma)} \gamma_\sigma(y)\, dy < 0, \tag{A.2}$$

since $\gamma_\sigma > 0$ on $(0, 1)$ and $\widehat{q} \in (0, 1)$. Hence $0 < S\beta_S'(\sigma) < O|\beta_O'(\sigma)|$. □

Proof of Proposition 2(i). By Lemma 4, the posterior gap $\beta_S(\sigma) - \beta_O(\sigma)$ is differentiable with positive derivative, hence continuous and strictly increasing on $(0, \infty)$.

Limits. All integrands below are bounded by 1, so dominated convergence applies under Assumption 2(iv). The mass $m_0(x, \sigma)$ defined in Appendix A.1 is strictly decreasing in x, with $m_0(\widehat{q}(\sigma), \sigma) = S$. As $\sigma \downarrow 0$, $m_0(x, \sigma) \to \max\{F(q_O) - F(x), 0\}$ at each x; since $F' > 0$, this limit crosses S only at $F^{-1}(1 - O - S) \in (0, q_O)$, from above. Applying the pointwise limit at $F^{-1}(1 - O - S) \mp \epsilon$ therefore traps $\widehat{q}(\sigma)$ within ϵ of $F^{-1}(1 - O - S)$ for all sufficiently small σ, so $\widehat{q}(\sigma) \to F^{-1}(1 - O - S)$. Passing to the limit in (4)–(5), the sponsored pool converges to $\{F^{-1}(1 - O - S) \le q \le q_O\}$ and the organic pool to $\{q > q_O\}$, so

$$\lim_{\sigma \downarrow 0} [\beta_S(\sigma) - \beta_O(\sigma)] = \mathbb{E}[q \mid F^{-1}(1 - O - S) \le q \le q_O] - \mathbb{E}[q \mid q > q_O] < 0,$$

strictly since $F' > 0$: the first mean lies strictly below q_O and the second strictly above. As $\sigma \to \infty$, $p \to O$ pointwise, so $m_0(x, \sigma) \to (1 - O)[1 - F(x)]$ and the same bracketing gives $\widehat{q}(\sigma) \to F^{-1}(1 - S/(1 - O)) \in (0, 1)$; organic-display status becomes independent of quality, so

$\beta_O(\sigma) \to \bar{q}$ and

$$\lim_{\sigma\to\infty}[\beta_S(\sigma) - \beta_O(\sigma)] = \mathbb{E}[q \mid q \geq F^{-1}(1 - S/(1-O))] - \bar{q} > 0,$$

strictly since $F' > 0$ places positive mass below the interior truncation point.

Existence, uniqueness, and interiority. A continuous, strictly increasing function with limits of opposite sign has a unique zero $\widehat{\sigma}(S) \in (0, \infty)$, and the posterior gap has the sign of $\sigma - \widehat{\sigma}(S)$; restricted to $I(S)$, this is the claimed equivalence. Since the gap depends on (F, p, O, S) alone, the threshold does not depend on κ or Φ. For interiority: whenever $\beta_S(\sigma) \leq \beta_O(\sigma)$, the left side of (NS) is nonpositive while $f^*(\sigma) > 0$ by (6), because $\kappa > 0$, Φ is strictly increasing, and $\beta_S(\sigma) > \widehat{q}(\sigma) > 0$; hence $(0, \widehat{\sigma}(S)] \subset I(S)$. At $\widehat{\sigma}(S)$, (NS) thus holds with strict slack, and both sides are continuous in σ, so it also holds on a right-neighborhood of $\widehat{\sigma}(S)$. Therefore $\widehat{\sigma}(S) \in \operatorname{int} I(S)$. □

Proof of Proposition 2(ii). Fix $S_2 > S_1$. Capacity gives $\widehat{q}(\sigma, S_2) < \widehat{q}(\sigma, S_1)$, so the additional advertisers have quality below $\beta_S(\sigma, S_1)$, the mean quality of sponsored sellers at ad load S_1. Hence $\beta_S(\sigma, S_2) < \beta_S(\sigma, S_1)$, while β_O is unchanged. At $\sigma = \widehat{\sigma}(S_1)$, the gap for S_2 is therefore negative. Since that gap strictly increases in σ, its unique zero satisfies $\widehat{\sigma}(S_2) > \widehat{\sigma}(S_1)$. □

Proof of Proposition 2(iii). By Proposition B.2(i), every fulfilled-expectations equilibrium (Definition B.1) is either the selective-bypass equilibrium or a switching equilibrium; by Proposition B.2(iii), every switching equilibrium has $\beta_S > \beta_O$.

Fix $\sigma > \widehat{\sigma}$. Any switching equilibrium at σ carries a premium; and if the selective-bypass equilibrium exists at σ, then $\sigma \in I(S)$ by Proposition B.2(ii), and part (i) gives $\beta_S(\sigma) > \beta_O(\sigma)$. Hence every equilibrium at σ has $\beta_S > \beta_O$.

Fix $\sigma < \widehat{\sigma}$. Since $(0, \widehat{\sigma}] \subset I(S)$ (established in the proof of part (i)), the selective-bypass equilibrium exists at σ by Proposition B.2(ii), and part (i) gives $\beta_S(\sigma) < \beta_O(\sigma)$; any other equilibrium at σ is switching and has $\beta_S > \beta_O$. □

Proof of Proposition 3. Capacity implies the upper-tail identity

$$(O+S)\mathcal{M}_\sigma([x,1]) = \min\left\{1 - F(x),\ O + S - \int_0^x p(q,\sigma)\, dF(q)\right\}.$$

Indeed, the second expression applies for $x \leq \widehat{q}(\sigma)$, where the sponsored tail has mass S; above the cutoff, every seller is displayed and the first expression applies. For each $x \in (0,1)$, the second

expression has derivative $-\gamma_\sigma(x) < 0$, whereas the first is constant in σ. Thus all upper tails weakly decrease. If $\sigma_2 > \sigma_1$, then $\widehat{q}(\sigma_2) > \widehat{q}(\sigma_1) > 0$, so the second expression applies at both noise levels for $0 < x < \widehat{q}(\sigma_1)$ and

$$(O+S)[\mathcal{M}_{\sigma_1}([x,1]) - \mathcal{M}_{\sigma_2}([x,1])] = \int_{\sigma_1}^{\sigma_2} \gamma_s(x)\, ds > 0.$$

This proves strict first-order stochastic dominance. □

A.4 Platform Choice and Disclosure (Section 5)

Proof of Lemma 5. Let $\lambda_2 > \lambda_1$ and let σ_i be an optimum at λ_i. Write $W_i = W(\sigma_i)$ and $R_i = R(\sigma_i)$. If $W_2 < W_1$, optimality at λ_1 and (PS) give

$$\frac{N_{\lambda_2}(W_2)R_2}{N_{\lambda_2}(W_1)R_1} < \frac{N_{\lambda_1}(W_2)R_2}{N_{\lambda_1}(W_1)R_1} \leq 1,$$

contradicting optimality at λ_2. Hence $W_2 \geq W_1$. Optimality at λ_1 then implies $R_1 \geq R_2$, and strict increase of R yields $\sigma_2 \leq \sigma_1$. Under no disclosure, the same argument gives $W_{ND}(\sigma_2) \geq W_{ND}(\sigma_1)$; the strict decrease of W_{ND} established below under "Click surplus and algorithm noise" again yields $\sigma_2 \leq \sigma_1$. □

Proof of Proposition 4. (i) Participation scales all bids by the same positive factor. Per participant, an undisplayed seller bids $\kappa(q)\Phi(\bar{q}_{\text{disp}})$ under no disclosure, while a seller granted organic display gains nothing from advertising. The same capacity equation determines $\widehat{q}$, with $f_{ND} = \kappa(\widehat{q})\Phi(\bar{q}_{\text{disp}})$. Since $\beta_S - \bar{q}_{\text{disp}} = O(\beta_S - \beta_O)/(O+S)$, the sign of $f^* - f_{ND}$ equals the sign of $\sigma - \widehat{\sigma}$.

(ii) Strict convexity of ω and Jensen's inequality give $O\omega(\beta_O) + S\omega(\beta_S) \geq (O+S)\omega(\bar{q}_{\text{disp}})$, with equality iff $\beta_S = \beta_O$. Since N_λ is strictly increasing, the same ordering holds for participation.

(iii) The positive objective ratio equals (8); the common factor $S\kappa(\widehat{q})$ cancels. Apply Lemma A.2(ii).

For the example following the proposition, temporarily write $x = \beta_S/\beta_O$; Lemma 4 gives $x' > 0$. The objective ratio is proportional to $x(Sx^2+O)^\lambda/(Sx+O)^{1+2\lambda}$. Its logarithmic derivative with respect to x is

$$\frac{O\{O + S[(1+2\lambda)x^2 - 2\lambda x]\}}{x(Sx^2+O)(Sx+O)}.$$

The quadratic in braces has minimum $O - S\lambda^2/(1+2\lambda)$, proving the sufficient bound. □

Click surplus and algorithm noise. Equation (A.2) implies the strict derivative $\bar{q}'_{\text{disp}} = (O\beta'_O + S\beta'_S)/(O+S) < 0$. Hence

$$W'_{ND} = (O+S)\Phi(\bar{q}_{\text{disp}})\bar{q}'_{\text{disp}} < 0.$$

Under disclosure,

$$W'_D = \Phi(\beta_O)(O+S)\bar{q}'_{\text{disp}} + S[\Phi(\beta_S) - \Phi(\beta_O)]\beta'_S.$$

The first term is strictly negative. The second is nonpositive when $\beta_S \leq \beta_O$, so $W'_D < 0$ in the penalty region and at $\sigma = \widehat{\sigma}$. In the premium region, the second term is positive and the sign of W'_D need not be negative.

The limit of high participation sensitivity. Suppose $N_\lambda(W_2)/N_\lambda(W_1) \to 0$ as $\lambda \to \infty$ for every $W_1 > W_2 > 0$. Fix $d \in \{D, ND\}$, write $W_d^{\max} = \max_{\sigma\in\Sigma} W_d(\sigma)$, and choose $\bar{\sigma}_d$ attaining this maximum. Continuity and compactness give $R_d^{\max} = \max_{\sigma\in\Sigma} R_d(\sigma) < \infty$, with $R_d(\bar{\sigma}_d) > 0$, where $R_D = R$. For any $0 < \epsilon < W_d^{\max}$, an optimum with $W_d(\sigma_d^*(\lambda)) \leq W_d^{\max} - \epsilon$ would satisfy

$$1 \leq \frac{N_\lambda(W_d(\sigma_d^*))R_d(\sigma_d^*)}{N_\lambda(W_d^{\max})R_d(\bar{\sigma}_d)} \leq \frac{R_d^{\max}}{R_d(\bar{\sigma}_d)} \frac{N_\lambda(W_d^{\max} - \epsilon)}{N_\lambda(W_d^{\max})} \longrightarrow 0,$$

a contradiction for sufficiently large λ. Thus $W_d(\sigma_d^*(\lambda)) \to W_d^{\max}$ in each regime. If $W_D^{\max} > W_{ND}^{\max}$, disclosure therefore raises click surplus for all sufficiently large λ. The strict decrease of W_{ND} gives $W_{ND}^{\max} = W_{ND}(\sigma_{\min})$. When $\sigma_{\min} \neq \widehat{\sigma}$, Proposition 4(ii) then gives $W_D^{\max} \geq W_D(\sigma_{\min}) > W_{ND}(\sigma_{\min}) = W_{ND}^{\max}$.

Proof of Remark 1. If $\sigma_{ND}^* > \widehat{\sigma}$, disclosure raises revenue per participant and participation at σ_{ND}^*. Its optimized profit is therefore strictly higher. Under the second condition, no disclosure yields strictly higher profit at σ_D^*, and hence also after optimization. □

Proof of Corollary 1. Write $M_r = (w_O c_O^r + w_S c_S^r)^{1/r}$ for the weighted power mean of the two disclosure click rates, for $r > 0$. All click rates are positive.

Threshold. Under $M_1 < c_{ND} < \max\{c_O, c_S\}$ the two label-specific rates are distinct. The strict power mean inequality makes M_r continuous and strictly increasing in $r > 0$, with limit $\max\{c_O, c_S\}$ as $r \to \infty$. Hence there is a unique $r^* > 1$ with $M_{r^*} = c_{ND}$.

Welfare. Define $\omega_r(y) = \omega(\Phi^{-1}(y^{1/r}))$. For $\theta = \Phi^{-1}(y^{1/r})$, direct differentiation gives

$$\omega_r''(y) = \frac{y^{2/r-2}}{r^2\Phi'(\theta)}\big[\mathcal{R}_\Phi(\theta) - r\big].$$

Thus ω_r is convex between c_O^r and c_S^r whenever $\mathcal{R}_\Phi \geq r$ throughout the interval between the two disclosure posteriors. Jensen's inequality then yields

$$w_O\omega(\Phi^{-1}(c_O)) + w_S\omega(\Phi^{-1}(c_S)) \geq \omega(\Phi^{-1}(M_r)).$$

At $r = r^*$ the right side equals $\omega(\Phi^{-1}(c_{ND}))$, proving $\Delta W \geq 0$ after multiplication by $O + S$. Since $\omega \circ \Phi^{-1}$ is increasing, $\Delta W \geq 0$ also holds whenever $M_r \geq c_{ND}$. Under the same smoothness assumptions, log-concavity gives $\Phi\Phi'' \leq (\Phi')^2$ and hence $\mathcal{R}_\Phi \geq 1$; concavity gives $\Phi'' \leq 0$ and hence $\mathcal{R}_\Phi \geq 2$. Taking $r = 1$ and $r = 2$, respectively, proves the two criteria stated after the corollary. □

B Equilibrium Correspondences

This appendix completes the equilibrium analysis. Appendix B.1 characterizes the full binary-quality correspondence and formalizes the self-reinforcing switching incentives in the three partial-switching equilibria omitted from Section 3.3. Appendix B.2 characterizes the sponsored-listing auction with continuous seller quality and supplies the taxonomy behind Proposition 2(iii), whose proof appears in Appendix A.3.

B.1 Binary-Quality Model

In the binary-quality model, write the ex ante cell masses

$$\begin{aligned} w_{H0} &\equiv \mu[1 - p_H(\sigma)], \quad & w_{L0} &\equiv (1-\mu)[1 - p_L(\sigma)], \\ w_{H1} &\equiv \mu p_H(\sigma), & w_{L1} &\equiv (1-\mu)p_L(\sigma). \end{aligned}$$

The organic-status posteriors are

$$\beta_{v=0}(\sigma) = \frac{w_{H0}}{w_{H0} + w_{L0}}, \qquad \beta_{v=1}(\sigma) = \frac{w_{H1}}{w_{H1} + w_{L1}},$$

so that $\beta_{v=0} < \mu < \beta_{v=1}$. Bayes' rule gives

$$\beta_S = \frac{w_{H0}\alpha_{H,0} + w_{H1}\alpha_{H,1}}{w_{H0}\alpha_{H,0} + w_{L0}\alpha_{L,0} + w_{H1}\alpha_{H,1} + w_{L1}\alpha_{L,1}}, \qquad \beta_O = \frac{w_{H1}(1-\alpha_{H,1})}{w_{H1}(1-\alpha_{H,1}) + w_{L1}(1-\alpha_{L,1})}$$

whenever the relevant pool has positive mass; advertising incentives are given by the payoff differences $\Delta_{t,v}$ defined in Section 3.1. In addition to $b = \Phi^{-1}(f/\kappa_L)$, write

$$b_{HL} \equiv \Phi^{-1}\left(\frac{f(\kappa_H - \kappa_L)}{\kappa_H \kappa_L}\right), \qquad b_H \equiv \Phi^{-1}\left(1 - \frac{f}{\kappa_H}\right)$$

for the organic beliefs that make $(H,1)$ indifferent when $\beta_S = b$ and $\beta_S = 1$, respectively. The taxonomy below groups strategy profiles with the same sponsored composition and seller payoffs.

B.1.1 The Complete Generic On-Path Taxonomy

In every nondegenerate on-path equilibrium $\beta_S > 0$. Lemma 1 then implies that $(H,0)$ has the highest gross return to advertising and, away from the knife edge $f = \kappa_H$, advertises with probability one whenever sponsorship is on path. Setting $\alpha_{H,0} = 1$, the taxonomy is driven by the two potentially marginal cells $(L,0)$ and $(H,1)$. Table B.1 summarizes the eight cases; the derivations are given in the proof of Proposition B.1 below.

B.1.2 Exhaustiveness

Proposition B.1 *(Complete binary-quality taxonomy)***.** *Every generic nondegenerate on-path equilibrium belongs to exactly one case in Table B.1. Here, generic excludes equality boundaries between cases, and on-path means that both labels occur with positive mass.*

Proof. Highest-return cell. Both labels are on path and $f > 0$. Some seller advertises, so $\Delta_{H,0} \geq 0$. If $\alpha_{H,0} < 1$, optimality gives $\kappa_H \Phi(\beta_S) = f$. Both low-quality cells then strictly abstain, forcing $\beta_S = 1$ and $f = \kappa_H$, an excluded knife edge. Thus $\alpha_{H,0} = 1$.

No switching: $\alpha_{H,1} = 0$. Optimality of $(H,1)$ implies $\alpha_{L,1} = 0$: when the label gap is positive its return is strictly lower, and otherwise it is nonpositive. The choices $\alpha_{L,0} = 1$, $0 < \alpha_{L,0} < 1$, and $\alpha_{L,0} = 0$ give cases 1–3. In case 1, $(L,0)$ is the binding cell, giving $f \leq \kappa_L \Phi(\beta_{v=0})$ and the penalty $\beta_{v=0} < \beta_{v=1}$. Case 2 is Lemma 2. In case 3, $\beta_S = 1$; the three relevant incentives are $f \leq \kappa_H$, $f \geq \kappa_L$, and $f \geq \kappa_H[1 - \Phi(\beta_{v=1})]$, exactly the table's conditions.

Table B.1: Equilibrium correspondence of the binary-quality model at a fixed σ.

Complete generic on-path equilibrium correspondence

Case	Advertising rates	(β_S, β_O)	Existence conditions	Label sign
1. Pure bypass	$(1, 1, 0, 0)$	$(\beta_{v=0},\ \beta_{v=1})$	$0 < f \leq \kappa_L \Phi(\beta_{v=0})$	Penalty
2. Selective bypass	$(1, \alpha_{L,0}, 0, 0)$	$(b,\ \beta_{v=1})$	$\kappa_L \Phi(\beta_{v=0}) < f < \kappa_L$ and $\beta_{v=1} \geq b_{HL}$	Flips at $\widehat{\sigma}(f)$ for $f > \kappa_L \Phi(\mu)$
3. Denied-high only	$(1, 0, 0, 0)$	$(1,\ \beta_{v=1})$	$\max\{\kappa_L,\ \kappa_H[1 - \Phi(\beta_{v=1})]\} \leq f \leq \kappa_H$	Premium
4. Denied-high with partial switching	$(1, 0, \alpha_{H,1}, 0)$	$(1,\ b_H)$	$\max\{\kappa_L,\ \kappa_H[1 - \Phi(\beta_{v=1})]\} < f < \kappa_H$	Premium
5. Double mixing	$(1, \alpha_{L,0}, \alpha_{H,1}, 0)$	$(b,\ b_{HL})$	$0 < f < \kappa_L$, $b_{HL} < \beta_{v=1}$, and $\alpha_{L,0} < 1$	Premium
6. All-denied with partial switching	$(1, 1, \alpha_{H,1}, 0)$	See proof	$0 < f < \kappa_H \Phi(\bar{\beta}_S)$ and $\kappa_L \Phi(\beta_S) \geq f$	Premium
7. Pure high-quality separation	$(1, 0, 1, 0)$	$(1,\ 0)$	$\kappa_L \leq f \leq \kappa_H$	Premium
8. All-high with low-quality mixing	$(1, \cdot, 1, \cdot)$	$(b,\ 0)$	$\kappa_L \Phi(\mu) < f < \kappa_L$	Premium

Notes. Profiles list $(\alpha_{H,0}, \alpha_{L,0}, \alpha_{H,1}, \alpha_{L,1})$ in that order. In case 8, the dots allow different advertising rates for low-quality sellers granted and denied organic display, while their total advertising mass $m_L = w_{L0}\alpha_{L,0} + w_{L1}\alpha_{L,1}$ is fixed. These rates yield the same label posteriors and seller payoffs. The terms b, b_{HL}, and b_H are defined above, and $\bar{\beta}_S = \mu/[\mu + (1 - \mu)(1 - p_L(\sigma))]$.

Partial switching: $0 < \alpha_{H,1} < 1$. Indifference gives $\Phi(\beta_S) - \Phi(\beta_O) = f/\kappa_H > 0$, so $\Delta_{L,1} = (\kappa_L/\kappa_H - 1)f < 0$ and $\alpha_{L,1} = 0$. The choices $\alpha_{L,0} = 0$, $0 < \alpha_{L,0} < 1$, and $\alpha_{L,0} = 1$ give cases 4–6. Every such equilibrium carries a premium.

In case 4, $\beta_S = 1$ and $\beta_O = b_H$. Bayes' rule gives $\alpha_{H,1} = 1 - w_{L1}b_H/[w_{H1}(1 - b_H)]$. Interiority is equivalent to $0 < b_H < \beta_{v=1}$, or $\kappa_H[1 - \Phi(\beta_{v=1})] < f < \kappa_H$; $(L, 0)$ abstains when $f \geq \kappa_L$. Away from equality boundaries these are the table's conditions.

In case 5, the two indifference conditions pin $\beta_S = b$ and $\beta_O = b_{HL}$, with

$$\alpha_{H,1} = 1 - \frac{w_{L1}b_{HL}}{w_{H1}(1 - b_{HL})}, \qquad \alpha_{L,0} = \frac{(w_{H0} + w_{H1}\alpha_{H,1})(1 - b)}{w_{L0}b}.$$

The conditions $0 < f < \kappa_L$, $b_{HL} < \beta_{v=1}$, and $\alpha_{L,0} < 1$ are exactly those making both rates interior; $\alpha_{H,1} < 1$ and $\alpha_{L,0} > 0$ then follow. The remaining cells have strict incentives.

In case 6,

$$\beta_S = \frac{w_{H0} + w_{H1}\alpha_{H,1}}{w_{H0} + w_{L0} + w_{H1}\alpha_{H,1}}, \qquad \beta_O = \frac{w_{H1}(1 - \alpha_{H,1})}{w_{H1}(1 - \alpha_{H,1}) + w_{L1}}.$$

The gap $\Phi(\beta_S) - \Phi(\beta_O)$ strictly increases from $\Phi(\beta_{v=0}) - \Phi(\beta_{v=1}) < 0$ to $\Phi(\bar{\beta}_S)$. Thus the indifference condition has a unique interior solution precisely when $0 < f < \kappa_H\Phi(\bar{\beta}_S)$; $(L, 0)$ additionally requires $\kappa_L\Phi(\beta_S) \geq f$. For $0 < f \leq \kappa_L\Phi(\mu)$ this last condition is automatic: all sellers are displayed and $\beta_S > \beta_O$, hence $\beta_S > \mu$.

All high-quality sellers advertise: $\alpha_{H,1} = 1$. The nonempty organic pool is entirely low quality, so $\beta_O = 0$ and $\alpha_{L,1} < 1$. Both low-quality cells have the same return. If $m_L = 0$, the high and low incentives give $\kappa_L \leq f \leq \kappa_H$: case 7. If $m_L > 0$, some low-quality seller advertises while some remains organic, so indifference pins $\beta_S = b$ and $m_L = \mu(1 - b)/b$. The condition $0 < m_L < 1 - \mu$ is equivalent to $\kappa_L\Phi(\mu) < f < \kappa_L$: case 8. For a given m_L, different pairs of advertising rates $(\alpha_{L,0}, \alpha_{L,1})$ satisfying $w_{L0}\alpha_{L,0} + w_{L1}\alpha_{L,1} = m_L$ and $\alpha_{L,1} < 1$ yield the same posteriors and seller payoffs, although the mass of organic listings can differ. Both cases carry a premium.

These eight cases exhaust all generic on-path profiles. □

Two additional cases fall outside this generic on-path taxonomy and are treated separately. The no-advertising profile leaves the sponsored label off path and is discussed under D1 in Appendix B.1.4; and the profile with $\alpha_{H,1} = \alpha_{L,1} = 1$ empties the organic pool, leaving the organic label off path.

B.1.3 Self-Reinforcing High-Quality Switching

In cases 4–6 of Table B.1, high-quality sellers eligible for organic display mix between advertising and retaining organic display. These equilibria share a positive-feedback mechanism: when more of these sellers advertise, the expected quality of organic listings falls, making advertising more attractive. I formalize this mechanism by comparing nearby advertising profiles, holding the fee f and algorithm noise σ fixed and updating consumers' beliefs by Bayes' rule.

Lemma B.1 *(Self-reinforcing high-quality switching). Consider an equilibrium in the generic strict-inequality region of case 4, 5, or 6. For any sufficiently small increase or decrease in the advertising rate of $(H,1)$ sellers, there is a nearby feasible profile at which all other seller types continue to play best responses, while $(H,1)$ sellers strictly prefer advertising after an increase and strictly prefer organic display after a decrease.*

Proof. Let $x = \alpha_{H,1}$ and $y = \alpha_{L,0}$, and denote their equilibrium values by x^* and y^*. Keep $\alpha_{H,0} = 1$ and $\alpha_{L,1} = 0$. Bayes' rule gives

$$\beta_S(x,y) = \frac{w_{H0} + w_{H1}x}{w_{H0} + w_{H1}x + w_{L0}y}, \qquad \beta_O(x) = \frac{w_{H1}(1-x)}{w_{H1}(1-x) + w_{L1}}.$$

Thus, increasing x strictly lowers the organic posterior. The net gain from advertising for an $(H,1)$ seller is

$$\Delta_{H,1}(x,y) = \kappa_H\big[\Phi(\beta_S(x,y)) - \Phi(\beta_O(x))\big] - f.$$

Because these sellers mix at the equilibrium, $x^* \in (0,1)$ and $\Delta_{H,1}(x^*,y^*) = 0$.

Cases 4 and 6. Change x to $x^* + \varepsilon$, keeping $y = y^*$. In case 4, $y^* = 0$, so the sponsored posterior remains equal to one. In case 6, $y^* = 1$, so the sponsored posterior strictly increases with x. In both cases, the organic posterior strictly decreases with x. Since Φ is strictly increasing, $\Delta_{H,1}(x,y^*)$ strictly increases with x. It is therefore positive when $\varepsilon > 0$ and negative when $\varepsilon < 0$.

Case 5. Here, both x^* and y^* are interior. To preserve the indifference of $(L,0)$ sellers, adjust y along with x so that the sponsored posterior remains at its equilibrium value

$$b = \beta_S(x^*,y^*) = \Phi^{-1}(f/\kappa_L).$$

Specifically, choose

$$x^\varepsilon = x^* + \varepsilon, \qquad y^\varepsilon = y^* + \frac{w_{H1}(1-b)}{w_{L0}b}\,\varepsilon.$$

For sufficiently small $|\varepsilon|$, both rates remain interior and $\beta_S(x^\varepsilon, y^\varepsilon) = b$. Hence, $(L, 0)$ sellers remain indifferent. Using $(H, 1)$ indifference at the equilibrium (x^*, y^*),

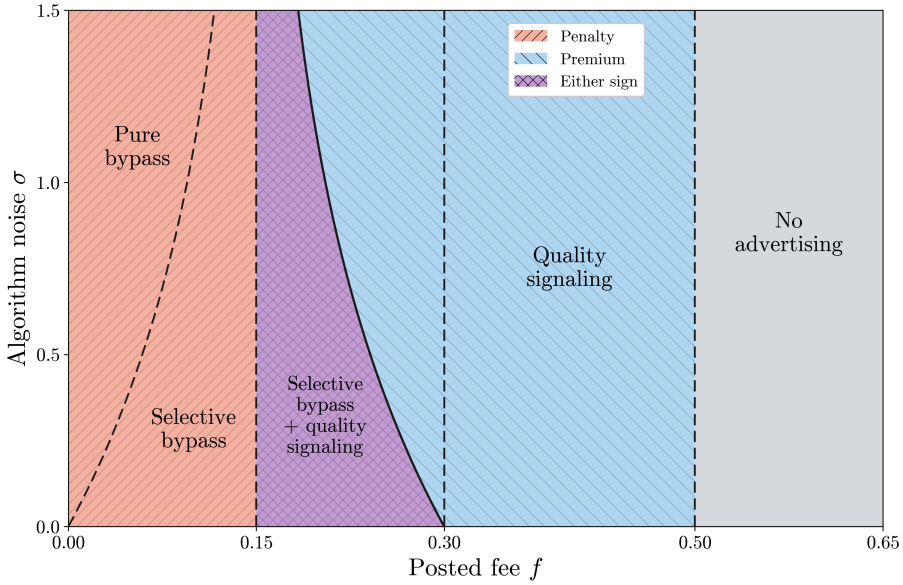
$$\Delta_{H,1}(x^\varepsilon, y^\varepsilon) = \kappa_H \big[\Phi(\beta_O(x^*)) - \Phi(\beta_O(x^* + \varepsilon))\big].$$

Because the organic posterior strictly decreases with x, this expression is positive when $\varepsilon > 0$ and negative when $\varepsilon < 0$.

In all three cases, the seller types playing pure strategies have strict incentives at (x^*, y^*). By continuity, their choices remain optimal for sufficiently small perturbations. In case 5, the other mixing type, $(L, 0)$, remains indifferent by construction. Thus, all types other than $(H, 1)$ continue to play best responses. □

Case 5 shows that the decline in organic quality alone is sufficient for this feedback: even when sponsored quality remains unchanged, additional high-quality switching makes advertising more attractive. Lemma B.1 describes this feedback; it imposes no equilibrium refinement and does not eliminate cases 4–6 from the correspondence.

Figure B.1: Label Inference: Binary Quality



Notes. Parameter values are $\mu = 0.5$, $\kappa_H = 0.5$, $\kappa_L = 0.3$, $\Phi(\beta) = \beta$, $p_H(\sigma) = \frac{1}{2}(1 + e^{-\sigma})$, and $p_L(\sigma) = \frac{1}{2}(1 - e^{-\sigma})$. Colors indicate penalties, premiums, or no advertising for the cases in Section 3.3; hatching marks where a penalty and a premium are both possible. The solid curve marks the penalty-to-premium threshold in selective bypass. Vertical dashed lines mark fee ranges; the dashed curve separates pure from selective bypass.

B.1.4 Multiplicity, Refinement, and Equilibrium Selection

Among the five cases in Section 3.3, multiplicity arises in two fee regions. The interval $\kappa_L\Phi(\mu) < f < \kappa_L$ can support both selective bypass (case 2) and all-high with low-quality mixing (case 8). Denied-high only (case 3) can coexist with pure high-quality separation (case 7) at higher fees. Only the first overlap can generate disagreement over the label sign: selective bypass may imply a penalty, whereas all-high with low-quality mixing implies a premium. Figure B.1 marks the parameters where both signs are possible.

The no-advertising profile can also be sustained by sufficiently pessimistic off-path beliefs, but it fails D1 (Cho and Kreps, 1987; Banks and Sobel, 1987): the $(H, 0)$ cell has the strongest incentive to deviate to advertising, so D1 assigns the deviation to the high-quality seller and hence sets $\beta_S = 1$, and the deviation is then profitable whenever $f < \kappa_H$. D1 therefore eliminates the no-advertising equilibrium for these fees but leaves the cases in Table B.1 unchanged. For $f > \kappa_H$, even the highest possible return to advertising is below the fee, giving the "No advertising" region in Figure B.1.

Appendix B.1.3 identifies self-reinforcing switching incentives in cases 4–6. I study selective bypass to isolate the interaction between bypassing and quality sorting, consistent with the evidence discussed in Section 4.1 that listings with greater organic strength are less likely to be advertised.

B.2 The Sponsored-Listing Auction

B.2.1 Equilibrium Definition and the Two-Cutoff Structure

Fix σ. The auction uses a uniform price and fills ad load S. An organic listing vacated by a sponsored buyer is not reassigned. Sellers choose whether to advertise deterministically, apart from a set of indifferent sellers with zero total mass. Write $m_1(x, \sigma) = \int_x^1 p(q, \sigma)\, dF(q)$ for the mass of $v = 1$ sellers above x.

Definition B.1 *(Fulfilled-expectations equilibrium).* *Fix σ. A fulfilled-expectations equilibrium of the sponsored-listing auction is a normalized sponsorship fee $f \geq 0$, a measurable purchase rule $a : [0, 1] \times \{0, 1\} \to \{0, 1\}$, and posteriors (β_S, β_O) such that:*

(a) Seller optimality. *Write $\Gamma_v(q) = \kappa(q)[\Phi(\beta_S) - v\Phi(\beta_O)]$ for the value of a sponsored listing to a seller with organic status v. Then $a(q, v) = 1$ implies $\Gamma_v(q) \geq f$ and $a(q, v) = 0$ implies $\Gamma_v(q) \leq f$, for a.e. (q, v).*

(b) Capacity. $\int_0^1 \left[a(q,0)\,(1-p(q,\sigma)) + a(q,1)\,p(q,\sigma)\right] dF(q) = S$.[14]

(c) Consistency. *β_S and β_O obey Bayes' rule on the sponsored pool $\{a=1\}$ and the organic pool $\{v=1,\, a=0\}$ whenever these have positive mass.*

Lemma B.2 *(Two-cutoff structure of auction equilibria). Fix σ and let (f, a, β_S, β_O) be an equilibrium. Then $f > 0$, the organic pool has positive mass with on-path posterior $\beta_O \in (0,1)$, and there exist cutoffs $0 < \hat{q}_0 < \hat{q}_1 \leq 1$ such that, except possibly for a set of sellers with zero mass, $a(q,0) = \mathbf{1}\{q \geq \hat{q}_0\}$ and $a(q,1) = \mathbf{1}\{q \geq \hat{q}_1\}$, and the following hold:*

(i) Capacity: *$m_0(\hat{q}_0, \sigma) + m_1(\hat{q}_1, \sigma) = S$; in particular, $v = 0$ buyers have strictly positive mass;*

(ii) $v = 0$ margin: *$\kappa(\hat{q}_0)\,\Phi(\beta_S) = f$;*

(iii) $v = 1$ margin: *if $\hat{q}_1 < 1$ then $\kappa(\hat{q}_1)\big[\Phi(\beta_S) - \Phi(\beta_O)\big] = f$ and $\beta_S > \beta_O$; if $\hat{q}_1 = 1$ then $\kappa(1)\big[\Phi(\beta_S) - \Phi(\beta_O)\big] \leq f$, which is condition (NS);*

(iv) Posteriors:

$$\beta_S = \frac{1}{S}\left[\int_{\hat{q}_0}^1 q\,[1-p(q,\sigma)]\,dF(q) + \int_{\hat{q}_1}^1 q\,p(q,\sigma)\,dF(q)\right], \qquad \beta_O = \frac{\int_0^{\hat{q}_1} q\,p(q,\sigma)\,dF(q)}{\int_0^{\hat{q}_1} p(q,\sigma)\,dF(q)}.$$

Moreover $\hat{q}_0 \geq \widehat{q}(\sigma)$, with equality if and only if $\hat{q}_1 = 1$.

Proof.

The proof uses throughout that F has a strictly positive density ($F' > 0$), so F is atomless and every nondegenerate subinterval of $[0,1]$ has positive F-mass.

Step 1 (positive price, the $v = 0$ cutoff, and (ii)). By (b) the sponsored pool has mass $S > 0$, so (c) gives $\beta_S > 0$ — the pool cannot consist only of sellers with $q = 0$, who have zero mass — and $\Phi(\beta_S) > 0$. Since $\Phi(\beta_O) \geq 0$ whatever its value, $\Gamma_1 \leq \Gamma_0$ pointwise. If $\kappa(0)\Phi(\beta_S) \geq f$, then $\Gamma_0(q) > f$ for every $q > 0$, so (a) forces $a(q,0) = 1$ a.e. and the sponsored mass is at least $1 - O > S$, violating (b); hence $f > \kappa(0)\Phi(\beta_S) > 0$. If $\kappa(1)\Phi(\beta_S) < f$, then $\Gamma_v \leq \Gamma_0 < f$ everywhere and the sponsored mass is $0 \neq S$; hence $f \leq \kappa(1)\Phi(\beta_S)$. By continuity and strict monotonicity of κ, the equation $\kappa(\hat{q}_0)\Phi(\beta_S) = f$ therefore has a unique solution $\hat{q}_0 \in (0,1]$, which is (ii), and (a) gives

[14] Allowing sponsored slots to remain unfilled would also admit an equilibrium in which no seller advertises: the belief $\beta_S = 0$ makes advertising generate no clicks. Requiring advertiser mass S excludes this outcome.

$a(q,0) = \mathbf{1}\{q \geq \hat{q}_0\}$ up to the F-null indifference set. If $\hat{q}_0 = 1$, then $f = \kappa(1)\Phi(\beta_S)$, so $\Gamma_v \leq \Gamma_0 < f$ on $q < 1$ and the sponsored pool is F-null, contradicting (b); hence $\hat{q}_0 < 1$ and the $v = 0$ buyer mass $m_0(\hat{q}_0, \sigma)$ is positive.

Step 2 (the organic pool and β_O). For a.e. $q < \hat{q}_0$, $\Gamma_1(q) \leq \Gamma_0(q) < f$ whatever the value of β_O, so (a) forces $a(q,1) = 0$. The organic pool thus contains a.e. $v = 1$ seller below $\hat{q}_0$ and has mass at least $\int_0^{\hat{q}_0} p\, dF > 0$, so Bayes' rule in (c) determines $\beta_O \in (0,1)$ because $q \in (0,1)$ F-a.s.

Step 3 (the $v = 1$ cutoff, (iii), and $\hat{q}_0 < \hat{q}_1$). If $\Gamma_1(1) \leq f$, then $\Gamma_1(q) < f$ for every $q < 1$ (also when $\beta_S \leq \beta_O$, since $f > 0$), so a.e. $v = 1$ seller abstains: $\hat{q}_1 = 1$ and (iii) is exactly (NS). Otherwise $\Gamma_1(1) > f > 0$ forces $\beta_S > \beta_O$; Γ_1 is then continuous and strictly increasing, with $\Gamma_1(0) \leq \Gamma_0(0) < f$. There is a unique $\hat{q}_1 \in (0,1)$ with $\Gamma_1(\hat{q}_1) = f$, and (a) gives $a(q,1) = \mathbf{1}\{q \geq \hat{q}_1\}$ up to the F-null indifference set, proving (iii). If $\hat{q}_1 = 1$, Step 1 gives $\hat{q}_0 < \hat{q}_1$; otherwise $\Phi(\beta_O) > 0$ gives $\Gamma_0(\hat{q}_1) > \Gamma_1(\hat{q}_1) = f = \Gamma_0(\hat{q}_0)$, so $\hat{q}_1 > \hat{q}_0$ by strict monotonicity.

Step 4 ((i), (iv), and comparison with $\widehat{q}$). Substituting the two cutoff rules into (b) gives $m_0(\hat{q}_0, \sigma) + m_1(\hat{q}_1, \sigma) = S$, which with $m_0(\hat{q}_0, \sigma) > 0$ from Step 1 is (i); substituting them into (c) gives the formulas in (iv). Finally, $m_0(\hat{q}_0, \sigma) = S - m_1(\hat{q}_1, \sigma) \leq S = m_0(\widehat{q}(\sigma), \sigma)$ and $m_0(\cdot, \sigma)$ is strictly decreasing, so $\hat{q}_0 \geq \widehat{q}(\sigma)$, with equality iff $m_1(\hat{q}_1, \sigma) = 0$ iff $\hat{q}_1 = 1$ (using $p > 0$, $F' > 0$). □

Call $\hat{q}_1 = 1$ selective bypass and $\hat{q}_1 < 1$ switching. Selective bypass has $\hat{q}_0 = \widehat{q}$ and the allocation of Lemma 3. Under switching, the highest-quality $v = 1$ sellers buy sponsored placement and $\beta_S > \beta_O$. Because vacated organic listings are not reassigned, the undisplayed mass becomes $1 - O - S + m_1(\hat{q}_1, \sigma)$.

B.2.2 Completeness and Existence

Parametrize candidate equilibria by the $v = 1$ cutoff. Define $q_T = F^{-1}(1 - S)$, the quality cutoff above which sellers have total mass S. For $x \in [q_T, 1]$, let $q_0(x)$ solve $m_0(q_0(x), \sigma) = S - m_1(x, \sigma)$. Write $\beta_S(x; \sigma), \beta_O(x; \sigma)$ for the posteriors in Lemma B.2(iv) at cutoffs $(q_0(x), x)$, holding σ fixed. Since $m_0(q_T, \sigma) + m_1(q_T, \sigma) = S$, the map q_0 is continuous and strictly decreasing, with $q_0(q_T) = q_T$ and $q_0(1) = \widehat{q}(\sigma)$. Thus $0 < q_0(x) < x$ for $q_T < x \leq 1$. Conversely, capacity and $q_0 < x$ imply $1 - F(x) < S$, so every equilibrium cutoff belongs to this interval.

Define the marginal $v = 1$ seller's excess willingness to pay by

$$\Delta_1(x) = [\kappa(x) - \kappa(q_0(x))]\Phi(\beta_S(x;\sigma)) - \kappa(x)\Phi(\beta_O(x;\sigma)).$$

The function Δ_1 is continuous on $[q_T, 1]$, both candidate posteriors are positive, and

$$\Delta_1(q_T) = -\kappa(q_T)\Phi(\beta_O(q_T;\sigma)) < 0, \qquad \Delta_1(1) \leq 0 \iff \text{(NS)}.$$

Interior zeros of Δ_1 characterize switching equilibria:

Proposition B.2 *(Continuum equilibrium taxonomy and existence). Fix σ. Then:*

(i) (Completeness.) *Every equilibrium is either the selective-bypass equilibrium or a switching equilibrium.*

(ii) (Selective bypass.) *The selective-bypass equilibrium exists if and only if (NS) holds at the selective-bypass posteriors (equivalently, $\sigma \in I(S)$ or $\Delta_1(1) \leq 0$).*

(iii) (Switching.) *$\hat{q}_1 \in (q_T, 1)$ supports a switching equilibrium if and only if $\Delta_1(\hat{q}_1) = 0$; the associated normalized sponsorship fee is $f = \kappa(q_0(\hat{q}_1))\Phi(\beta_S(\hat{q}_1;\sigma))$. Every switching equilibrium carries a sponsorship premium, $\beta_S > \beta_O$.*

(iv) (Existence for all σ.) *If (NS) fails, a switching equilibrium exists. Hence the auction market clears at every $\sigma \in (0, \infty)$.*

Proof. Lemma B.2 establishes (i). When the $v = 1$ cutoff equals one, capacity uniquely gives $\hat{q}_0 = \widehat{q}$, and the remaining incentive constraint is (NS); Lemma 3 therefore establishes (ii).

For (iii), every switching equilibrium has $q_T < \hat{q}_1 < 1$ by the cutoff ordering and capacity. Its two marginal indifference conditions give $\Delta_1(\hat{q}_1) = 0$ and the stated price. Conversely, let $x \in (q_T, 1)$ satisfy $\Delta_1(x) = 0$ and set $f = \kappa(q_0(x))\Phi(\beta_S(x;\sigma)) > 0$. Then

$$\kappa(x)[\Phi(\beta_S(x;\sigma)) - \Phi(\beta_O(x;\sigma))] = f > 0.$$

Both willingness-to-pay schedules are strictly increasing and equal f at their respective cutoffs, so both purchase rules are optimal. Capacity and Bayes consistency hold by construction. The displayed identity also gives $\beta_S(x;\sigma) > \beta_O(x;\sigma)$.

Finally, if (NS) fails, then $\Delta_1(q_T) < 0 < \Delta_1(1)$. Continuity and the intermediate value theorem give an interior zero, establishing (iv). □

B.2.3 Coexistence and Uniqueness

Selective-bypass and switching equilibria can coexist for the same primitives. Switching depends on the strength of the signaling incentive: how rapidly the value per click κ increases with quality. Define the mean quality among sellers with $v = 1$ and $q \le \widehat{q}(\sigma)$ by

$$\underline{\beta}_O(\sigma) \;\equiv\; \mathbb{E}[q \mid v = 1,\; q \le \widehat{q}(\sigma)] = \frac{\int_0^{\widehat{q}(\sigma)} q\, p(q,\sigma)\, dF(q)}{\int_0^{\widehat{q}(\sigma)} p(q,\sigma)\, dF(q)}.$$

By Proposition B.2(ii)–(iii), if $\Delta_1(1) \le 0$ and Δ_1 has a zero in $(q_T, 1)$, selective bypass coexists with switching. The sign of $\beta_S - \beta_O$ in selective bypass is given by Proposition 2, while every switching equilibrium carries a premium.

Proposition B.3 *(A sufficient condition for uniqueness). Fix σ. If $\kappa(1)\big[1 - \Phi(\underline{\beta}_O(\sigma))\big] < \kappa(\widehat{q}(\sigma))$, then no switching equilibrium exists at σ, (NS) holds strictly, and the selective-bypass equilibrium is the unique fulfilled-expectations equilibrium at σ.*

Proof. The conditional mean $\mathbb{E}[q \mid v = 1,\; q \le x]$ increases with x. In any switching equilibrium, Lemma B.2 therefore gives $\beta_O \ge \underline{\beta}_O$ and $\hat{q}_1 > \hat{q}_0 \ge \widehat{q}$. The marginal conditions imply

$$\kappa(\hat{q}_0) = \kappa(\hat{q}_1)\left[1 - \frac{\Phi(\beta_O)}{\Phi(\beta_S)}\right] \le \kappa(1)[1 - \Phi(\underline{\beta}_O)] < \kappa(\widehat{q}) \le \kappa(\hat{q}_0),$$

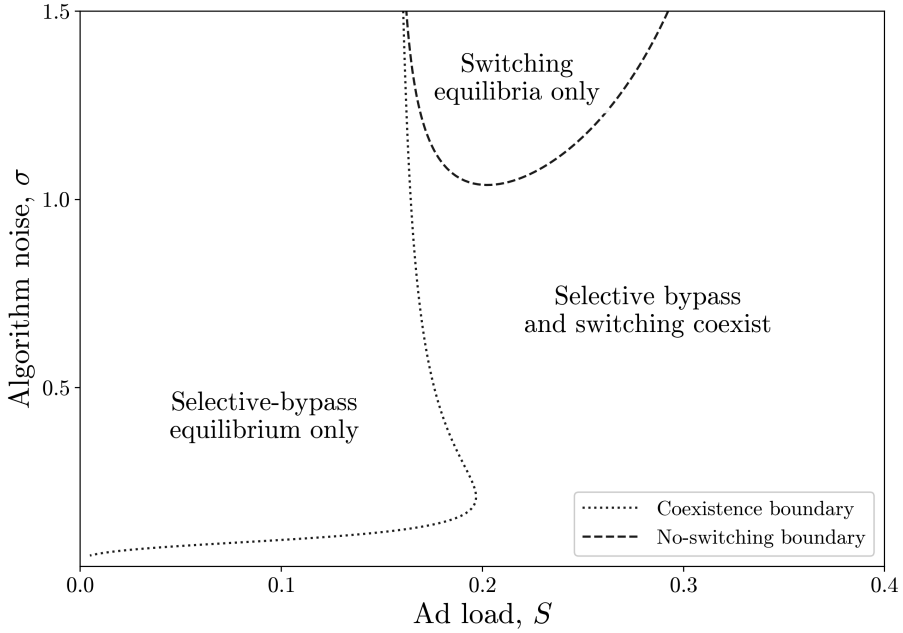
a contradiction; the weak inequality uses $\Phi(\beta_S) \le 1$. At the selective-bypass posteriors the same bound gives

$$\kappa(1)\left[1 - \frac{\Phi(\beta_O)}{\Phi(\beta_S)}\right] \le \kappa(1)[1 - \Phi(\underline{\beta}_O)] < \kappa(\widehat{q}),$$

so (NS) holds strictly. Completeness then establishes uniqueness. □

The condition bounds the relative growth of κ above the selective-bypass cutoff: $\kappa(1)/\kappa(\widehat{q}) < [1 - \Phi(\underline{\beta}_O)]^{-1}$. Both $\widehat{q}$ and $\underline{\beta}_O$ are determined by (F, p, S, O), so the condition can be checked without solving for a switching equilibrium. If this condition holds at $\sigma < \widehat{\sigma}$, every equilibrium carries a sponsorship penalty. If it holds for all $0 < \sigma < \widehat{\sigma}$, every equilibrium carries a penalty below $\widehat{\sigma}$ and, by Proposition 2(iii), a premium above $\widehat{\sigma}$. The condition is sufficient but not necessary

Figure B.2: Existence regions of selective-bypass and switching equilibria in the (S, σ) plane.



Notes. Same specification as Figure 1: $q \sim \text{Unif}[0,1]$, $\varepsilon \sim N(0,1)$, $O = 0.6$, $\Phi(\beta) = \beta$, $\kappa(q) = 0.02 + 0.98\,q^2$. The dotted curve separates unique selective bypass from coexistence with switching equilibria. The dashed curve marks the loss of selective bypass; beyond it, only switching equilibria exist.

because it uses a uniform bound on candidate switching equilibria.

If the condition in Proposition B.3 holds at every $\sigma \in \Sigma$, selective bypass is unique throughout the policy interval. Under the assumptions of Section 5, Lemma 5 then gives weakly decreasing optimal noise in λ under each disclosure regime. With the additional condition in Proposition 4(iii), optimal noise is also weakly higher under disclosure.

Figure B.2 maps the existence regions in the (S, σ) plane under the illustrative specification of Figure 1.

Scope. The comparative statics in Sections 4.2–5 are established in the selective-bypass equilibrium, $\sigma \in I(S)$. In a switching equilibrium, both the displayed mass and the allocation rule change, so the effects of noise on revenue and displayed quality need not extend to switching equilibria.

C Foundations and Extensions

This appendix provides the noisy-score foundation of the organic-listing allocation rule, a click–purchase microfoundation, an extension to personalized organic display, the participation micro-

foundation, and an extension to transaction commissions.

C.1 Noisy-Score Foundation

This subsection verifies all four parts of Assumption 2 for the noisy-score rule in Example 1. Let Ψ denote the CDF of the score noise ε, with density Ψ'.

Lemma C.1 *(Noisy-score foundation).* *The organic-display rule in Example 1 satisfies Assumption 2. Its capacity cutoff $\bar{s}(\sigma)$ is C^1, and the crossing point in part (iii) is $q^\dagger(\sigma) = \bar{s}(\sigma) - \sigma\bar{s}'(\sigma) \in (0,1)$.*

Proof. Write $p(q,\sigma) = 1 - \Psi((\bar{s}(\sigma) - q)/\sigma)$. For each $\sigma > 0$, the capacity equation has a unique cutoff because its left side is continuous and strictly decreasing in $\bar{s}$, with limits one and zero. Its derivative with respect to $\bar{s}$ is $-\sigma^{-1}\int_0^1 \Psi'((\bar{s}-q)/\sigma)\,dF(q) < 0$. Local boundedness of the continuous derivatives permits differentiation under the integral, so the implicit function theorem gives $\bar{s} \in C^1$ and hence $p \in C^1$.

(i) The strictly positive noise density gives $0 < p < 1$ and $p_q(q,\sigma) = \Psi'((\bar{s} - q)/\sigma)/\sigma > 0$.

(ii) The cutoff satisfies $\int_0^1 p(q,\sigma)\,dF(q) = O$ by construction.

(iii) Differentiating the display probability yields

$$p_\sigma(q,\sigma) = \frac{\Psi'((\bar{s}(\sigma) - q)/\sigma)}{\sigma^2}\big[\bar{s}(\sigma) - q - \sigma\bar{s}'(\sigma)\big].$$

Differentiating capacity identifies the crossing point:

$$q^\dagger(\sigma) = \bar{s} - \sigma\bar{s}' = \frac{\int_0^1 q\,\Psi'((\bar{s} - q)/\sigma)\,dF(q)}{\int_0^1 \Psi'((\bar{s} - q)/\sigma)\,dF(q)} \in (0,1).$$

Thus p_σ is strictly positive below $q^\dagger$ and strictly negative above it.

(iv) As $\sigma \downarrow 0$, dominated convergence gives $\int_0^1 [1 - \Psi((x-q)/\sigma)]\,dF(q) \to 1 - F(x)$ for each fixed x. Evaluating at $q_O \pm \epsilon$ brackets the capacity cutoff and gives $\bar{s}(\sigma) \to q_O$, hence $p(q,\sigma) \to \mathbf{1}\{q > q_O\}$ for a.e. q. For the other limit, $q \in [0,1]$ gives

$$1 - \Psi(\bar{s}/\sigma) \le O \le 1 - \Psi((\bar{s} - 1)/\sigma), \qquad \Psi^{-1}(1-O) \le \bar{s}/\sigma \le \Psi^{-1}(1-O) + 1/\sigma.$$

Thus $(\bar{s} - q)/\sigma \to \Psi^{-1}(1-O)$ uniformly in q as $\sigma \to \infty$, and continuity of Ψ gives $p(q,\sigma) \to O$ uniformly. □

C.2 A Click–Purchase Microfoundation

Section 2 treats the click probability $\Phi(\beta_\ell)$ and the per-click value $\kappa(q)$ as primitives. Both can be derived from a consumer who first clicks and then decides whether to purchase, with quality measured in units of her expected value from clicking.

Let i identify a consumer and j a seller. Seller j has product quality $z_j \in [\underline{z}, \bar{z}]$, which is common to all consumers. Across sellers, z_j has a continuous, strictly positive density. Consumer i's match value with seller j is $\xi_{ij} \sim H$, where H has a continuous density h, full support on $\mathbb{R}$, and $\mathbb{E}[\max\{\xi_{ij}, 0\}] < \infty$. Match values are independent across pairs (i, j) and of z_j and organic display. Here v_j denotes seller j's organic-display realization, which is the same for all consumers as in Section 2. The match-value distribution H is held fixed when algorithm noise σ changes.

The consumer observes the label $\ell \in \{S, O\}$ and her click cost before clicking. When participation is introduced, both are observed after the participation decision, as in Section 5.1. A click reveals net purchase utility $z_j + \xi_{ij}$. She purchases iff it is nonnegative and otherwise obtains zero.

Consumer side. For a seller with $z_j = z$, the expected value from clicking before paying the click cost is $J(z) = \mathbb{E}[\max\{z + \xi_{ij}, 0\}]$. It is continuous and strictly increasing, with derivative $J'(z) = 1 - H(-z) \in (0, 1)$. Normalize this expected value to define the quality q used in Section 2:

$$q \equiv \frac{J(z) - J(\underline{z})}{J(\bar{z}) - J(\underline{z})} \in [0, 1].$$

This is a strictly increasing bijection; write $z(q)$ for its inverse, the value of z corresponding to q. It induces a distribution F for q with continuous, strictly positive density, as assumed in Section 4.

Since match values are independent of organic display, the expected value from clicking a listing with label ℓ is $J(\underline{z}) + [J(\bar{z}) - J(\underline{z})]\beta_\ell$, where $\beta_\ell = \mathbb{E}[q_j \mid \ell]$ is the posterior mean quality from Section 2. Before normalization, let consumer i's click cost be

$$J(\underline{z}) + [J(\bar{z}) - J(\underline{z})]\theta_i, \qquad \theta_i \sim \Phi \text{ on } [0, 1],$$

independently of qualities, match values, and organic display. The consumer clicks iff $\theta_i \leq \beta_\ell$, with probability $\Phi(\beta_\ell)$. Thus the normalized click cost θ_i and quality q recover both the click decision and expected click surplus $\omega(\beta_\ell)$ in Section 5. Surplus in the original units equals $[J(\bar{z}) - J(\underline{z})]\omega(\beta_\ell)$;

outside options are expressed in the same normalized units when participation is introduced. If $z_j \in \{\underline{z}, \bar{z}\}$, the same normalization gives $q \in \{0, 1\}$ and $\beta_\ell = \mathbb{P}(q_j = 1 \mid \ell)$, recovering the posterior in Section 3.

Seller side. The seller's per-purchase margin is a constant $\varrho > 0$. For a seller with $z_j = z(q)$, the purchase probability after a click is $1 - H(-z(q))$. Hence the per-click value is

$$\kappa(q) = \varrho[1 - H(-z(q))].$$

It is continuous and strictly increasing in q, giving the per-click value assumed in Section 4 and $\kappa_H > \kappa_L$ in Section 3. Since H is held fixed, $\kappa(q)$ does not change with algorithm noise σ.

C.3 Personalized Search Results

Organic search results can use information from a consumer's query or browsing history to identify products that better match the consumer's tastes. This extension captures such personalization by allowing organic display to depend on a noisy signal of consumer–seller match value. Here the platform observes seller quality exactly, and algorithm noise affects its signal of match value. Sellers bid before match values and signal errors are realized, without observing their organic-display realizations.

The symbols z_j, H, and ξ_{ij} retain their roles from Appendix C.2, but the utility normalization differs. The binary purchase payoff specified below makes quality equal to a randomly drawn consumer's purchase probability; Appendix C.2 instead normalizes expected click value under continuous purchase utility.

Match value and quality. There is a unit mass of atomless sellers and a unit mass of potential consumers. The subscript i identifies a consumer, and j identifies a seller. Seller j has quality q_j, drawn from a distribution F on $[0, 1]$ with a continuous, positive density. Consumer i's match value with seller j is $\xi_{ij} \sim H$, independently across pairs (i, j) and of quality. The match value has mean zero and a log-concave density h, continuous and positive inside its interval of possible values. This class includes uniform and normal match values.

Let z_j denote the component of purchase utility common to all consumers facing seller j. The

sum $z_j + \xi_{ij}$ determines whether the product suits consumer i. Net purchase utility is normalized to one when this sum is nonnegative and to minus one otherwise; not purchasing gives zero. Clicking reveals which applies, so the consumer purchases exactly when $z_j + \xi_{ij} \geq 0$. For convenience, express the common component z_j in terms of quality q_j, writing $z_j = z(q_j)$ so that a randomly drawn consumer purchases with probability q_j after clicking:

$$z(q) = -H^{-1}(1-q), \qquad X_{ij} = \mathbf{1}\{z(q_j) + \xi_{ij} \geq 0\}, \qquad \mathbb{P}(X_{ij} = 1 \mid q_j) = q_j.$$

Because the consumer can decline to purchase after clicking, the realized value of a click before paying its cost is X_{ij}. Its expectation for a randomly drawn consumer is therefore q_j, as in Appendix C.2. The function $z(q)$ is defined for $0 < q < 1$; the endpoints have zero mass.

Organic display and sponsorship. Organic capacity $O \in (0,1)$ and ad load $S \in (0,1)$ are given. The platform observes seller quality q_j and a noisy signal of each consumer's match value,

$$\tilde{\xi}_{ij} = \xi_{ij} + \sigma\varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \Psi, \qquad \sigma > 0.$$

The parameter σ scales the signal error. The error distribution Ψ has a continuous, strictly positive density Ψ' on $\mathbb{R}$. Errors are independent across pairs (i,j) and of qualities and match values. The organic algorithm grants seller j display to consumer i when

$$v_{ij} = \mathbf{1}\{z(q_j) + \tilde{\xi}_{ij} \geq \bar{s}(\sigma)\} = 1.$$

v_{ij} has the same meaning as v in Section 2, but now depends on both the seller and the consumer. The algorithm applies a cutoff to a noisy observation of the same expression that determines X_{ij}. The cutoff is chosen so that organic display would have expected mass O per consumer in the absence of sponsorship. Sponsored sellers are removed from organic results without replacement. Consumers do not observe q_j, ξ_{ij}, or $\tilde{\xi}_{ij}$ before clicking.

Sellers know their quality, the organic algorithm, and both capacities. Before match values and signal errors are realized, they bid in a uniform-price auction for sponsorship. The highest bids of mass S win and pay the same sponsorship fee f. Each winner is shown to every consumer as sponsored and receives no organic listing. A nonwinner is shown organically to consumer i exactly

when $v_{ij} = 1$. Unlike in Section 2, sellers therefore compare sponsorship with expected organic profit across consumers rather than conditioning their bids on realized organic display.

Consumer decisions and purchase probabilities. Consumer i pays a click cost $\theta_i \sim \Phi$ on $[0, 1]$, independently of qualities, match values, and the signal errors in the organic algorithm. As in Section 2, each consumer observes only the listing's label and its click cost before clicking, and evaluates listings independently. Assume Φ has a continuous, positive derivative on $(0, 1)$.

After clicking, the consumer purchases exactly when $X_{ij} = 1$. The probability of purchase differs between sponsored and organic listings because only organic display selects consumers by match value. For a seller of quality q, averaging over the match value gives

$$\begin{aligned} p(q,\sigma) &= \mathbb{P}(v_{ij} = 1 \mid q_j = q) = \int_{\mathbb{R}} \left[1 - \Psi\left(\frac{\bar{s}(\sigma) - z(q) - \xi}{\sigma}\right)\right] h(\xi)\, d\xi, \\ C(q,\sigma) &= \mathbb{P}(v_{ij} = 1, X_{ij} = 1 \mid q_j = q) = \int_{-z(q)}^{\infty} \left[1 - \Psi\left(\frac{\bar{s}(\sigma) - z(q) - \xi}{\sigma}\right)\right] h(\xi)\, d\xi. \end{aligned} \tag{C.1}$$

The cutoff condition is $\int_0^1 p(q,\sigma)\, dF(q) = O$. As in Section 2, $p(q,\sigma)$ is the organic-display probability, here for a given consumer. The additional function $C(q,\sigma)$ is the joint probability of organic display and a purchase if the consumer clicks. For a sponsored seller of quality q, display does not select consumers by match value. Its purchase probability conditional on a click is therefore q. For an organic seller of the same quality, only consumers with $v_{ij} = 1$ see the listing, and the purchase probability conditional on a click is $C(q,\sigma)/p(q,\sigma)$. Clicking itself does not further select match values within either group: consumers know only the label and their independently drawn click cost before clicking. Because the signal error ε_{ij} has positive density on $\mathbb{R}$, $0 < p(q,\sigma) < 1$ for every $0 < q < 1$. Since the algorithm favors higher match values, $C(q,\sigma)/p(q,\sigma) > q$.

Consumer inference. Let A contain the qualities of sellers who advertise, with $\int_A dF(q) = S$, and let $A^c = [0, 1] \setminus A$ contain the other qualities. Write $O_A = \int_{A^c} p(q,\sigma)\, dF(q)$ for the mass of organic listings per consumer after sellers with qualities in A advertise. For a displayed listing, retain $\ell \in \{S, O\}$ for its label, with S denoting sponsored and O organic, as in Section 2. The posterior mean quality is β_ℓ. Let b_ℓ be the click value, the consumer's expected value from clicking,

which here equals the purchase probability after clicking:

$$\beta_S = \frac{1}{S}\int_A q\,dF(q), \quad \beta_O = \frac{1}{O_A}\int_{A^c} q\,p(q,\sigma)\,dF(q),$$
$$b_S = \beta_S, \qquad\qquad b_O = \frac{1}{O_A}\int_{A^c} C(q,\sigma)\,dF(q) = \beta_O + \tau.$$

The difference

$$\tau = \frac{1}{O_A}\int_{A^c} [C(q,\sigma) - q\,p(q,\sigma)]\,dF(q) > 0$$

arises because, within each quality, consumers who receive organic display are more likely to purchase after clicking. The consumer clicks a displayed listing with label ℓ when $\theta_i \leq b_\ell$, giving click probability $\Phi(b_\ell)$. A sponsorship penalty occurs when $b_S < b_O$, equivalently when $\beta_S - \beta_O < \tau$. Thus, when $0 < \beta_S - \beta_O < \tau$, sponsored listings have higher posterior mean quality but a lower click probability conditional on display.

Sponsorship bypasses both quality screening and match-based selection into organic display. It extends a seller's reach to consumers who would not see its organic listing, while giving up the favorable match selection associated with organic exposure.

Seller payoffs and advertising decisions. The seller's per-purchase margin is a constant $\varrho > 0$. Its value per sponsored click is ϱq, while its value per organic click also reflects selection on match value. Taking b_S, b_O, and f as given, a seller's expected sponsored and organic profits are $\varrho q\Phi(b_S) - f$ and $\varrho C(q,\sigma)\Phi(b_O)$, respectively, per potential consumer. The sponsored per-click return ϱq plays the role of the increasing $\kappa(q)$ in Section 2: higher-quality sellers have more to gain from being clicked. Organic profit now depends on both how often a seller is displayed and how likely the consumers who see it are to purchase. The seller's willingness to pay for sponsorship, written $\Gamma(q)$ with the dependence on σ, b_S, and b_O suppressed, is

$$\begin{aligned}\Gamma(q) &= \varrho[q\Phi(b_S) - C(q,\sigma)\Phi(b_O)] \\ &= \varrho[q - C(q,\sigma)]\Phi(b_S) + \varrho C(q,\sigma)[\Phi(b_S) - \Phi(b_O)].\end{aligned} \tag{C.2}$$

The first term values purchases from consumers who would not receive organic display. The second term values the change in click probability among consumers who would both receive organic display and purchase after clicking.

Personalization makes the organic profit that sellers give up more valuable. To see this, keep the same advertisers and the same organic-display probability for each quality, but suppose that organic listings are assigned to consumers independently of match value. An organic click then leads to a purchase with probability q, so $C(q,\sigma)$ becomes $q\,p(q,\sigma)$ and the organic click value becomes β_O. Willingness to pay for sponsorship would be $\varrho[q\Phi(\beta_S) - q\,p(q,\sigma)\Phi(\beta_O)]$. With personalization, organic display instead reaches consumers more likely to purchase, and consumers are more willing to click organic listings: $C(q,\sigma) > q\,p(q,\sigma)$ and $b_O > \beta_O$. Both effects raise organic profit, so willingness to pay for sponsorship is lower under personalization than under independent assignment.

Under the personalized algorithm, sellers choose whether to advertise by comparing $\Gamma(q)$ with the common sponsorship fee f. Sellers with $\Gamma(q) > f$ advertise and those with $\Gamma(q) < f$ remain organic; sellers with $\Gamma(q) = f$ can be rationed to fill capacity. In equilibrium, the resulting advertiser set A has mass $\int_A dF(q) = S$, and the click values b_S and b_O used to compute bids equal those implied by A through consumer inference. Equivalently, $\{q : \Gamma(q) > f\} \subseteq A \subseteq \{q : \Gamma(q) \geq f\}$, except possibly for a set of sellers with zero mass.

Higher quality raises the gross return to sponsored clicks, but it also raises the organic profit forgone by advertising. Log-concavity of match values makes $C(q,\sigma)$ convex in quality. At fixed click values, sponsored revenue is linear in quality and organic profit is convex, so willingness to pay for sponsorship is concave.

Lemma C.2. *In every equilibrium with $f > 0$, the sellers who advertise have qualities in one interval.*

Proof. Substituting $\delta = \xi + z(q)$ in (C.1) and using $z'(q) = 1/h(-z(q))$ gives

$$\frac{\partial C(q,\sigma)}{\partial q} = 1 - \Psi\left(\frac{\bar{s}(\sigma)}{\sigma}\right) + \int_0^\infty \frac{1}{\sigma}\Psi'\left(\frac{\bar{s}(\sigma) - \delta}{\sigma}\right)\frac{h(\delta - z(q))}{h(-z(q))}\,d\delta.$$

For $\delta > 0$, the density ratio is nondecreasing in $z(q)$ by log-concavity. Thus $\partial C/\partial q$ is nondecreasing in q, so $C(q,\sigma)$ is convex in q and $\Gamma(q)$, a linear function of q minus $\varrho\Phi(b_O)C(q,\sigma)$, is concave when b_S and b_O are held fixed. Its upper level sets are therefore intervals. It remains to rule out a positive-mass tie at $f > 0$. A flat interval of Γ would make $\partial C/\partial q$ constant there. Since $\Psi' > 0$ and each density ratio in the integral is nondecreasing, each ratio must be constant throughout that interval. Log-concavity then forces h to be exponential on the corresponding right tail: otherwise

some positive shift δ would give a strict change in the ratio. On such a tail $C(q,\sigma)$ is proportional to q, so a constant Γ can only equal zero. Thus ties at a positive fee have zero mass, and advertisers form one interval except possibly for a set of sellers with zero mass. □

Write this interval as $[q_-, q_+]$, where the subscripts $-$ and $+$ denote the lowest and highest qualities among advertisers. The upper bound can be $q_+ = 1$. When $0 < q_- < q_+ < 1$, the highest-quality sellers retain organic display, as in the selective-bypass equilibrium of Section 4.

Comparative statics of algorithm noise. Noise affects organic click value both by changing which consumers see each seller and by changing which sellers advertise. In the general specification, the cutoff also adjusts to maintain organic capacity, so the direction of the change in organic purchase opportunities is not straightforward. To make these effects tractable, consider the following symmetric specification, in which the organic cutoff remains fixed as noise changes.

$$q_j \sim \text{Unif}[0,1], \qquad \xi_{ij} \sim \text{Unif}[-1/2, 1/2], \qquad \varepsilon_{ij} \sim N(0,1), \qquad O = 1/2.$$

Here Ψ is the standard normal CDF. Then $z(q_j) = q_j - 1/2$, and $X_{ij} = 1$ iff $q_j + \xi_{ij} \geq 1/2$. Symmetry of $z(q_j) + \xi_{ij} + \sigma\varepsilon_{ij}$ about zero gives $\bar{s}(\sigma) = 0$ at every noise level. With $u = q - 1/2 + \xi$, the two probabilities become

$$p(q,\sigma) = \int_{q-1}^{q} \Psi(u/\sigma)\, du, \qquad C(q,\sigma) = \int_{0}^{q} \Psi(u/\sigma)\, du.$$

Consider $0 < q_- < q_+ < 1$. Uniformly distributed qualities and the given ad load S imply $q_+ - q_- = S$. Hence write $q_- = \bar{q}_A - S/2$ and $q_+ = \bar{q}_A + S/2$, where $\bar{q}_A = b_S$ is mean quality among advertisers. In equilibrium, the two marginal advertisers at q_- and q_+ must have the same willingness to pay for sponsorship. To express this condition, write $\Delta C = C(q_+,\sigma) - C(q_-,\sigma) = \int_{q_-}^{q_+} \Psi(q/\sigma)\, dq$ for the difference in C between these sellers, and define

$$\mathcal{E}(\bar{q}_A, \sigma) = \Phi(\bar{q}_A) - \frac{\Delta C}{S}\, \Phi\big(b_O(\bar{q}_A, \sigma)\big).$$

Then $\Gamma(q_+) - \Gamma(q_-) = \varrho S \mathcal{E}$, so equilibrium requires $\mathcal{E} = 0$; by concavity of Γ, this condition is also sufficient. The value $b_O(\bar{q}_A, \sigma)$ is computed using sellers outside $[\bar{q}_A - S/2, \bar{q}_A + S/2]$. Raising $\bar{q}_A$

increases both q_- and q_+ while preserving advertiser mass S.

Under this specification, greater noise lowers organic click value at a fixed advertiser set. Under the slope condition below, it also shifts the advertising interval toward higher qualities.

Proposition C.1 *(Selection with personalized organic display).* *Under the preceding distributional assumptions and $O = 1/2$, consider an equilibrium with $f > 0$, $0 < q_- < q_+ < 1$, and $\partial\mathcal{E}/\partial\bar{q}_A < 0$. A marginal increase in algorithm noise raises q_-, q_+, and b_S, and lowers b_O. Moreover,*

$$\frac{d(b_S - b_O)}{d\sigma} = \underbrace{-\frac{\partial b_O}{\partial \sigma}}_{\text{holding } A \text{ fixed}} + \underbrace{\left(1 - \frac{\partial b_O}{\partial \bar{q}_A}\right)\frac{d\bar{q}_A}{d\sigma}}_{\text{change in who advertises}} > 0, \tag{C.3}$$

and both terms are strictly positive. The partial derivative with respect to σ holds $\bar{q}_A$ fixed; the partial derivative with respect to $\bar{q}_A$ holds σ fixed and adjusts both bounds to keep mass S.

Proof. With $\bar{s} = 0$, $C(q,\sigma) = \int_0^q \Psi(u/\sigma)\,du$ decreases in σ and $p(q,\sigma) - C(q,\sigma) = \int_{q-1}^0 \Psi(u/\sigma)\,du$ increases in σ for $0 < q < 1$: greater noise removes organic display from consumers who would purchase and extends it to consumers who would not. Holding A fixed, $b_O = \int_{A^c} C\,dF / \left[\int_{A^c} C\,dF + \int_{A^c}(p - C)\,dF\right]$ therefore falls, so $\partial b_O/\partial\sigma < 0$. Also,

$$\frac{\partial \Delta C}{\partial \sigma} = -\frac{1}{\sigma^2}\int_{q_-}^{q_+} q\,\Psi'(q/\sigma)\,dq < 0,$$
$$\frac{\partial \mathcal{E}}{\partial \sigma} = -\frac{1}{S}\frac{\partial \Delta C}{\partial \sigma}\Phi(b_O) - \frac{\Delta C}{S}\Phi'(b_O)\frac{\partial b_O}{\partial \sigma} > 0.$$

Thus $d\bar{q}_A/d\sigma = -(\partial\mathcal{E}/\partial\sigma)/(\partial\mathcal{E}/\partial\bar{q}_A) > 0$.

Next, define $\Delta p = p(q_+,\sigma) - p(q_-,\sigma)$. The derivatives $\partial C/\partial q = \Psi(q/\sigma)$ and $\partial p/\partial q = \Psi(q/\sigma) - \Psi((q-1)/\sigma)$ imply $\partial C/\partial q > \partial p/\partial q > 0$, so $\Delta C > \Delta p > 0$ and, since $b_O < 1$,

$$\frac{\partial b_O}{\partial \bar{q}_A} = -\frac{\Delta C - b_O \Delta p}{O_A} < 0.$$

Together with $b_S = \bar{q}_A$, these derivatives prove the result. □

The condition $\partial\mathcal{E}/\partial\bar{q}_A < 0$ concerns the relative bids of the two marginal advertisers. Shift the advertising interval slightly toward higher qualities, keeping its size fixed. Holding click values fixed isolates the direct effect of shifting the interval: the difference in organic profits between the two marginal sellers increases, lowering the upper seller's bid relative to the lower seller's. Consumer inference works in the opposite direction: the higher sponsored click value raises the bid at q_+ relative to that at q_-, and the lower organic click value reduces the opportunity cost of advertising by

more at q_+, where organic profit is larger. The condition requires the increase in the organic-profit difference to dominate these changes in click values.

Greater noise raises the bid at the old q_+ relative to the bid at the old q_-. Along the equilibrium branch, the interval shifts upward to restore equality between the two marginal bids.

Disclosure backfire. Personalization also affects the consequences of mandatory sponsorship disclosure when the platform chooses algorithm noise. As in Section 5.2, disclosure can improve consumers' click decisions while inducing the platform to choose a noisier algorithm. The following example uses a different quality distribution and organic capacity to show that backfire can also occur when equilibrium advertiser composition remains unchanged across disclosure regimes and throughout the feasible noise interval.

The distributions and capacities are reported in the figure notes. In equilibrium, the highest-quality sellers advertise under both regimes throughout the feasible noise interval: their qualities are $[q_-, 1]$, with $q_- = F^{-1}(1 - S)$. Let the platform choose noise with participation $N_\lambda(W_d)$, as in Section 5.2. Here W_d is ex ante expected click surplus per participant under regime $d \in \{D, ND\}$. Consumers decide whether to participate before observing click costs, match values, or listings; outside options are independent of these variables.

Let b_{ND} denote the pooled click value under no disclosure. Using $\omega(b) = \int_0^b \Phi(t)\,dt$ from Section 5, click values and surplus satisfy

$$b_{ND} = \frac{S b_S + O_A b_O}{S + O_A},$$
$$W_D = S\omega(b_S) + O_A\omega(b_O), \quad W_{ND} = (S + O_A)\omega(b_{ND}).$$

Write f_D and f_{ND} for sponsorship fees per participant, extending the fee f defined above for a unit mass of consumers:

$$f_D = \varrho[q_-\Phi(b_S) - C(q_-, \sigma)\Phi(b_O)],$$
$$f_{ND} = \varrho[q_- - C(q_-, \sigma)]\Phi(b_{ND}).$$

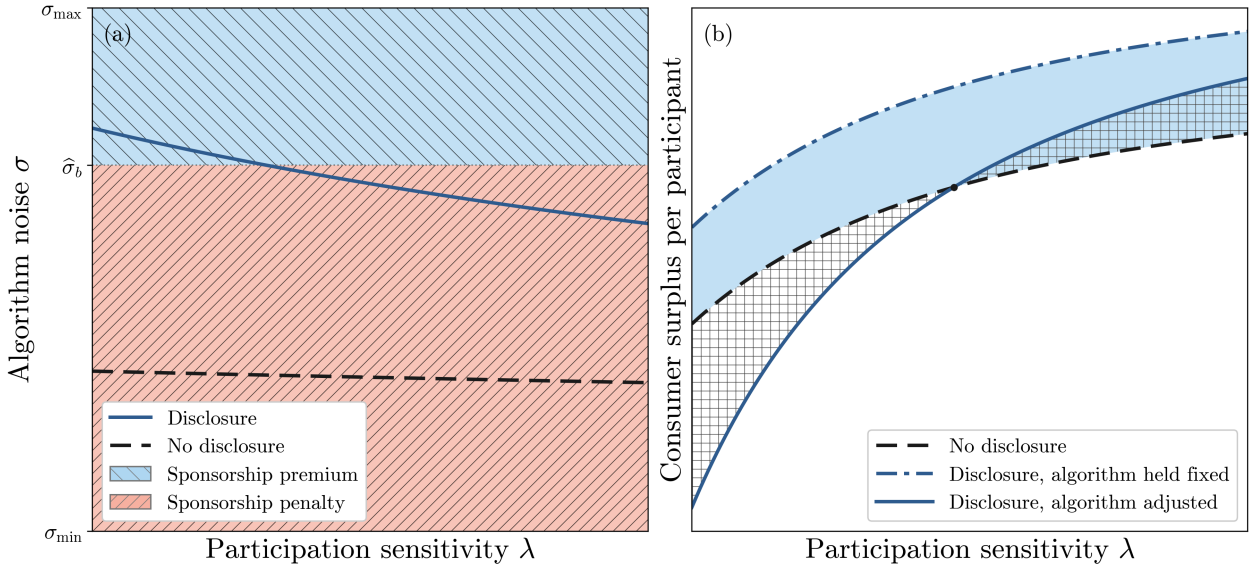
Participation scales bids by a common factor, so the actual fee is $N_\lambda(W_d) f_d$. The platform chooses $\sigma \in \Sigma$ to maximize $S N_\lambda(W_d(\sigma)) f_d(\sigma)$.

Greater noise reduces organic purchase opportunities, raising the value of sponsorship. Without disclosure, the decline in pooled click value also reduces sponsored clicks, limiting this revenue gain.

Disclosure can therefore induce greater noise whose cost to consumers outweighs the benefit of better-informed clicks.

Figure C.1 illustrates this tradeoff, as in Figure 2. Panel (a) focuses on optimal noise, while panel (b) uses a higher range of participation sensitivity to show where disclosure's effect on consumer surplus changes sign.

Figure C.1: Disclosure Backfire with Personalized Search Results



Notes. $q_j \sim \text{Beta}(10,3)$, $\xi_{ij} \sim \text{Unif}[-1/2, 1/2]$, $\varepsilon_{ij} \sim N(0,1)$, $O = 0.15$, $S = 0.05$, $\varrho = 1$, $\Phi(\theta) = \theta$, $\Sigma = [0.20, 0.75]$, and $N_\lambda(W) = (3W)^\lambda$. Panels (a) and (b) use $\lambda \in [0.68, 0.83]$ and $\lambda \in [3.5, 8]$, respectively. The top five percent of sellers advertise in both regimes; all plotted optima are interior, and the mandate binds throughout (b). In (a), $\widehat{\sigma}_b$ marks $b_S = b_O$; the penalty arises from organic selection on tastes despite $\beta_S > \beta_O$. Panel (b) uses the same line styles and shading as Figure 2(b).

In panel (a), greater participation sensitivity lowers optimal noise under both regimes. In panel (b), disclosure lowers consumer surplus when the loss from the algorithm adjustment exceeds the information gain; at higher participation sensitivity, the information gain dominates.

With personalized organic results, sponsorship exchanges match-selected organic exposure for access to all consumers. This preserves the tradeoff between sponsored reach and forgone organic profit, while allowing sponsored listings to have a lower click probability conditional on display even when advertisers have higher mean quality. The numerical example further shows that disclosure can reduce consumer surplus through the platform's choice of algorithm noise.

C.4 Participation and Consumer Welfare

Return to the selective-bypass equilibrium in Section 4, with $\sigma \in \Sigma \subset I(S)$. Consumers know σ and the disclosure regime $d \in \{D, ND\}$ before deciding whether to participate. They have not yet observed listings or click costs and anticipate the expected total click surplus $W_d(\sigma)$ defined in Section 5. Each consumer observes an outside option $o \sim G_\lambda$, independent of click costs and listing qualities, and participates iff $W_d(\sigma) \geq o$. Thus

$$N_\lambda(W) = G_\lambda(W).$$

Continuity and strict increase of G_λ on the relevant surplus range give parts (i) and (iii) of Assumption 3. Part (ii) requires the relative participation rate at a lower surplus to fall with λ. The timing and independence preserve the click-cost distribution Φ among participants.

For example, let the outside-option CDF be $G_\lambda(o) = (o/\bar{W})^\lambda$ on $[0, \bar{W}]$, where $\lambda > 0$ and $\bar{W}$ exceeds all relevant W_d. Then $N_\lambda(W_2)/N_\lambda(W_1) = (W_2/W_1)^\lambda$ strictly decreases with λ for $W_1 > W_2 > 0$. The parameter is the elasticity of participation with respect to expected click surplus. At $\lambda = 1$, participation is proportional to surplus. Figures 2 and C.1 use this family with $\bar{W} = 1/3$.

The same foundation connects click surplus per participant to aggregate consumer welfare. Suppose the outside option has finite expectation. Conditional on outside option o, the consumer's expected payoff is $\max\{W_d(\sigma), o\}$, including that outside option when not participating. At a fixed λ, the difference in aggregate welfare between the two optimized regimes is

$$\int [\max\{W_D(\sigma_D^*), o\} - \max\{W_{ND}(\sigma_{ND}^*), o\}]\, dG_\lambda(o) = \int_{W_{ND}(\sigma_{ND}^*)}^{W_D(\sigma_D^*)} G_\lambda(w)\, dw. \tag{C.4}$$

This identity follows by integrating the marginal gain $G_\lambda(w)$ from increasing click surplus. Since participation is positive, the aggregate welfare difference has the same sign as $W_D(\sigma_D^*) - W_{ND}(\sigma_{ND}^*)$, even though participation differs. Corollary 1 therefore also gives a sufficient condition for an aggregate consumer benefit. Its magnitude additionally depends on the outside-option distribution. This comparison holds λ fixed; varying λ changes the outside-option distribution as well as participation sensitivity. In the power family, aggregate surplus relative to taking the outside option is $W_d^{\lambda+1}/[(\lambda+1)\bar{W}^\lambda]$.

C.5 Transaction Commissions

Maintain Assumption 3 and the selective-bypass equilibrium on Σ. Let $K_d(\sigma) \geq 0$ be commission revenue per participating consumer in regime $d \in \{D, ND\}$, continuous in σ and independent of λ. Let $R_d(\sigma)$ denote sponsorship revenue per participating consumer under regime d, so that $R_D(\sigma) = R(\sigma)$. The platform maximizes

$$N_\lambda(W_d(\sigma))[R_d(\sigma) + K_d(\sigma)].$$

The extension treats $\kappa(q)$ as the seller's value per click net of any transaction commission, so K_d captures the platform's additional revenue while the auction allocation remains the one in Section 4, with κ understood net of commissions.

For each regime d and participation parameter λ, assume that $N_\lambda(W_d(\sigma))R_d(\sigma)$ and $N_\lambda(W_d(\sigma))[R_d(\sigma) + K_d(\sigma)]$ have unique maximizers on Σ, denoted by σ_d^* and $\sigma_{K,d}^*$, respectively.

Proposition C.2 *(Commissions).*

(i) *Optimal noise $\sigma_{K,d}^*$ weakly decreases with λ if either W_d is weakly decreasing or $R_d + K_d$ is strictly increasing in σ on Σ.*

(ii) *If $K_d(\sigma)/R_d(\sigma)$ is weakly decreasing in σ, then $\sigma_{K,d}^* \leq \sigma_d^*$.*

(iii) *If $\{N_\lambda(W_D(\sigma))/N_\lambda(W_{ND}(\sigma))\}[R(\sigma) + K_D(\sigma)]/[R_{ND}(\sigma) + K_{ND}(\sigma)]$ is weakly increasing in σ, then $\sigma_{K,D}^* \geq \sigma_{K,ND}^*$.*

Proof. For part (i), if W_d weakly decreases with noise, (PS) makes $N_{\lambda_2}(W_d)/N_{\lambda_1}(W_d)$ weakly decreasing in noise. Apply Lemma A.2(ii). If instead $R_d + K_d$ is strictly increasing, the proof of Lemma 5 applies with $R_d + K_d$ in place of R. For part (ii), participation is unchanged by commissions at a fixed algorithm, so the objective ratio is $1 + K_d/R_d$. For part (iii), the objective ratio is exactly the ratio in its condition. Apply Lemma A.2(ii) to both comparisons. □

The three parts concern different comparisons: part (i) varies participation sensitivity while holding commissions fixed; part (ii) introduces commission revenue while holding participation sensitivity and the disclosure regime fixed; part (iii) compares disclosure with no disclosure in the presence of commissions.

To illustrate these conditions, let the platform collect $\chi \in (0, \varrho)$ per purchase in Appendix C.2. With purchase probability $r(q) = 1 - H(-z(q))$, the seller's net per-click value becomes $\kappa(q) = (\varrho - \chi)r(q)$. This common scaling preserves the advertising allocation and consumer posteriors at a fixed σ, and the advertising-only optimum. Commission revenue per participant is

$$K_D = \chi[S\Phi(\beta_S)\mathbb{E}[r(q) \mid \ell = S] + O\Phi(\beta_O)\mathbb{E}[r(q) \mid \ell = O]],$$
$$K_{ND} = \chi(S + O)\Phi(\bar{q}_{\text{disp}}) \int r(q)\, \mathcal{M}_\sigma(dq).$$

Participation sensitivity: part (i). At a fixed commission rate, greater participation sensitivity leads to weakly less noise if click surplus decreases with noise. This condition holds under no disclosure and under disclosure on any policy interval contained in the penalty region; see "Click surplus and algorithm noise" in Appendix A.4. In the premium region, disclosure surplus need not decrease everywhere, but strictly increasing total revenue per participant provides the alternative condition in part (i). For the example above, if the revenue functions are continuously differentiable in noise with derivatives continuous in χ, then $R' > 0$ at $\chi = 0$ on compact Σ ensures that $R_D + K_D$ remains strictly increasing for sufficiently small χ.

The effect of commissions: part (ii). Now hold λ and the disclosure regime fixed. Commissions lower optimal noise when transaction revenue relative to advertising revenue, K_d/R_d, decreases with noise. Under no disclosure, the click probability cancels from K_{ND}/R_{ND}: displayed purchase probabilities fall with noise while the marginal advertiser's purchase probability rises. Under disclosure,

$$\frac{K_D}{R_D} = \frac{\chi}{(\varrho - \chi)Sr(\widehat{q})} \left[(S + O) \int r(q)\, \mathcal{M}_\sigma(dq) + O \left(\frac{\Phi(\beta_O)}{\Phi(\beta_S)} - 1 \right) \mathbb{E}[r(q) \mid \ell = O] \right].$$

The first term in brackets decreases by Proposition 3. In the penalty region the second is a product of positive, decreasing terms: organic listings have a declining purchase probability and click rate relative to sponsored listings. The denominator increases. Thus Proposition C.2(ii) applies under no disclosure and under disclosure on any policy interval contained in the penalty region.

Commissions also weakly reduce optimal noise in the premium region if match-value variation is sufficiently large that higher sponsored quality generates little increase in purchase probabilities.

Take $\xi_{ij} \sim N(0, \eta^2)$ and normally distributed score noise in Example 1, holding the distribution of z_j, its bounded support, and the normalized click-cost distribution fixed as η varies. Assume Φ' is continuous and positive on the relevant posterior range and $\Sigma \subset (0, \infty)$ is a fixed compact interval. As η grows, purchase probabilities approach $1/2$ and become insensitive to quality. More precisely, $r(q) = 1/2 + \mathcal{O}(\eta^{-1})$ and $r'(q) = \mathcal{O}(\eta^{-1})$ uniformly, so

$$\frac{\varrho - \chi}{\chi} \frac{K_D}{R_D} = 1 + \frac{O}{S} \frac{\Phi(\beta_O)}{\Phi(\beta_S)} + \mathcal{O}(\eta^{-1}),$$

uniformly on Σ, including first derivatives in σ. The limiting distribution of q is that of $(z_j - \underline{z})/(\bar{z} - \underline{z})$. Under normally distributed score noise, the derivative of the click-probability ratio is strictly negative and bounded away from zero on compact Σ at this limit. Consequently K_D/R_D decreases for sufficiently large η in both regions. Sponsored clicks then raise advertising and transaction revenue in nearly the same proportion, while lost organic clicks reduce transaction revenue alone.

The effect of disclosure: part (iii). With commissions present, hold λ fixed and compare the two disclosure regimes. Suppose that N_λ is continuously differentiable and that the profit ratio in (8) has a continuous, strictly positive derivative on Σ. This is the ratio at $\chi = 0$, including participation. Compactness then keeps its derivative positive for sufficiently small χ, so the platform still chooses weakly greater algorithm noise under disclosure than under no disclosure when commissions are sufficiently small.